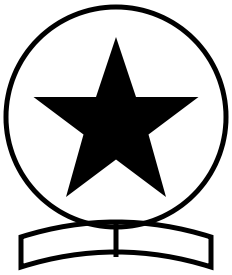
TEXAS STAAR



GRADE 8

Reading Language Arts & Mathematics

All-in-One Study & Practice Guide

8 Full-Length Practice Tests · Complete Skill Review · Deep Answer Explanations

Short & Extended Constructed Response Mastery · Score Diagnostics

100% aligned to the Texas Essential Knowledge and Skills (TEKS)

2027 EDITION

Andrian Ross

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All passages, questions, graphics, and practice materials in this book are **original works** created for instructional practice. They are designed to reflect the style, format, difficulty, and standards coverage of the STAAR Grade 8 assessment; they are **not** actual test questions and are **not** reproduced from any secure or released state testing material.

About This Edition

This 2027 Edition reflects the STAAR program **as of November 5, 2026**, including the online test format and the current question types, scoring, and TEKS blueprint. Under Texas House Bill 8 (2025), **spring 2027 is the final STAAR administration** for grades 3–8; a new assessment, the Student Success Tool, is scheduled to replace STAAR beginning in the 2027–2028 school year. This book is built to prepare students thoroughly for that final STAAR year.

While every effort has been made to ensure accuracy, test specifications can change. Families and educators should confirm current details on the official Texas Education Agency website.

Table of Contents

Page numbers are finalized during layout.

Getting Started

How to Use This Book

Inside the STAAR Grade 8 Test — The Final STAAR Year

How STAAR Is Scored

Item-Type Field Guide (How Online Questions Look on Paper)

Your Study Calendars (4-Week and 8-Week Plans)

Test-Day Game Plan

Part 1 · Mathematics

How Grade 8 Math Is Built (The Four Strands)

Grade 8 Mathematics Reference Sheet

Strand 1 — Numerical Representations and Relationships

Strand 2 — Computations and Algebraic Relationships

Strand 3 — Geometry and Measurement

Strand 4 — Data Analysis and Personal Financial Literacy

Process Standards in Action · Warm-Up Skill Drills

Math Practice Tests 1–4

Part 2 · Reading Language Arts

How Grade 8 RLA Is Built (Reading, Writing, and Written Responses)

Genre Guide · Reading Strategy Lessons · Revising and Editing Refresher

Short Constructed Response (SCR) Mastery

Extended Constructed Response (ECR) Mastery

RLA Practice Tests 1–4

Part 3 · Answer Keys & Explanations

Quick Answer Keys (All 8 Tests)

Mathematics — Answer Explanations (Tests 1–4)

Reading Language Arts — Answer Explanations (Tests 1–4)

SCR Scored Samples · ECR Scored Samples

Part 4 · Diagnose & Grow

Score-Tracking Sheets · Strand Diagnostic Charts

A Guide for Parents and Teachers

TEKS Coverage Index · Final Notes

How this book is organized: First you *learn* each tested skill, then you *practice* it in short drills, then you *simulate* the real test with eight full-length practice tests, and finally you *diagnose* your results to see exactly what to study next.

Welcome to Your STAAR Grade 8 Study Guide

Dear Student,

The STAAR Grade 8 tests in Reading Language Arts and Mathematics can feel like a lot — two subjects, many skills, a timed online screen, and even written essays. This book exists to make all of that feel familiar long before test day.

Here is the promise of this book: **everything inside is on the test, and nothing on the test is left out.** Every lesson, every practice question, and every writing prompt is tied directly to the Texas Essential Knowledge and Skills (TEKS) that STAAR measures at Grade 8. You will not waste a single minute on material that will not help you.

You will move through four stages:

1. **Learn** — short, clear lessons that teach every tested skill.
2. **Practice** — quick drills so each skill sticks.
3. **Simulate** — eight full-length practice tests that look and feel like the real thing.
4. **Diagnose** — answer explanations and charts that show you exactly what to review next.

Work at a steady pace, check every answer explanation (even for questions you got right), and use the diagnostic charts to spend your study time where it matters most. Confidence on test day comes from practice, and this book gives you plenty.

You can do this. Let's get started.

— *Andrian Ross*

> **A note for parents and teachers:** A full guide for supporting a student with this book — including how to read the diagnostic charts and how to coach the written responses — begins on the "A Guide for Parents and Teachers" page near the back.

How to Use This Book

This book works for three kinds of readers. Find the path that fits you.

Path A — The Student Studying on Your Own

1. **Start with "Getting Started."** Read how the test works, how it is scored, and how the online question types look on paper. About 30 minutes — and it saves you from surprises.
2. **Pick a study calendar.** Use the 4-Week Sprint if the test is close, or the 8-Week Plan if you have more time.
3. **Learn, then practice.** Read the skill lessons and do the warm-up drills before a full test.
4. **Take one full test at a time, under test-like conditions** — quiet room, no phone, timed.
5. **Review every explanation,** including for questions you answered correctly.
6. **Fill in your diagnostic chart** after each test, then study your weakest strand next.

Path B — The Parent or Tutor Helping a Student

- Use the **"A Guide for Parents and Teachers"** section at the back first.
- Sit with the student for the **first** practice test to model pacing, then let them work more independently.
- Focus praise on **effort and strategy**, not just the score. The tracking sheets make growth visible.

Path C — The Classroom Teacher

- Lessons and drills map one-to-one to the **TEKS**; the **TEKS Coverage Index** lists where every standard is taught and practiced.
- Each full-length test is blueprint-matched — usable as a benchmark or mock administration.
- Explanations include the TEKS code and *why each wrong choice is wrong*, useful for reteaching.

The Symbols You Will See

Symbol	Meaning
★	**Readiness standard** — tested heavily every year. Master these first.
○	**Supporting standard** — tested less often, but still fair game.

Key Idea	The one thing to remember from a lesson.
Watch Out	A common Grade 8 mistake to avoid.
Try It	A quick practice item to check yourself.
Strategy	A test-taking tip tied to the question in front of you.

Tip: Keep a pencil, scratch paper, and the Grade 8 Mathematics Reference Sheet nearby while you work. On the real test these tools are available to you — practice using them now.

Inside the STAAR Grade 8 Test

The Final STAAR Year

STAAR — the State of Texas Assessments of Academic Readiness — measures how well students have learned the **Texas Essential Knowledge and Skills (TEKS)** for their grade. Under Texas House Bill 8 (2025), **spring 2027 is the last STAAR administration** for grades 3–8. A new assessment called the Student Success Tool is scheduled to replace it in 2027–2028. This book prepares you fully for that final STAAR year.

A Computer-Based Test

STAAR is given **online**, on a computer. The real test includes question types beyond simple multiple choice — you might type an answer, choose from a drop-down menu, drag items into place, or build an equation. A paper book can't make you click and drag, so this book does the next best thing: it **shows every online question type on paper** and teaches you exactly how to answer it. The *thinking* is identical; only the clicking is different.

What Grade 8 Students Take

At Grade 8, students are assessed in **Reading Language Arts, Mathematics, Science, and Social Studies.** This book focuses on the two taken by every Grade 8 student every year: **Reading Language Arts and Mathematics.**

The Mathematics Test at a Glance

Feature	Grade 8 Mathematics
Total questions	**40**
Total points	**48** (32 one-point + 8 two-point questions)
Reporting categories (strands)	4 — Numerical; Computations & Algebraic; Geometry & Measurement; Data & Financial Literacy
Question types	Multiple choice, multiselect, inline choice, text entry/griddable, drag-and-drop, equation editor, graphing
Tools provided	**Calculator** and the **Grade 8 Mathematics Reference Sheet**

The Reading Language Arts Test at a Glance

Feature	Grade 8 Reading Language Arts
Total questions	**~45**
Total points	**56**
Reading section	~28 questions / ~30 points — includes **2 Short Constructed Responses** (2 pts each)
Writing section (Revising & Editing)	~17 questions / ~26 points — includes **1 Extended Constructed Response** (10 pts)
Passages	Literary and informational; single passages and a paired set
Tools provided	Online **dictionary/thesaurus**

How Much Time?

STAAR for grades 3–8 is given with a generous time window of **up to four hours**. It is **not a speed test** — most students finish comfortably. Still, practicing with a timer builds a steady, calm pace.

> **Watch Out:** Written responses carry real points — up to **14 of the 56 RLA points** come from the SCRs and ECR. Never leave them blank. A partial, organized response almost always earns more than nothing.

How STAAR Is Scored

From Raw Points to a Performance Level

Each question earns **raw points** (1 or 2 for most; the Extended Constructed Response is worth 10). Your total raw points convert to a **scale score**, which places you in one of four **performance levels**:

Performance Level	What It Means
Masters Grade Level	Strong command of the grade's content; ready for the next grade with little support.
Meets Grade Level	Solid understanding; likely to succeed next grade with some support.
Approaches Grade Level	Some understanding; likely needs targeted support next grade.
Did Not Meet Grade Level	Not enough understanding yet; needs significant support.

The goal for most students is **Meets Grade Level** or higher — and this book is built to get you there.

How Written Responses Are Scored

- **Short Constructed Response (SCR)** — a few sentences answering a specific question, scored **0 to 2 points**. Earn full credit by making a clear point and backing it with correct **evidence from the text**.
- **Extended Constructed Response (ECR)** — a full multi-paragraph essay, scored **0 to 10 points**. The rubric rewards a clear central idea or claim, organized paragraphs, specific supporting evidence, and correct language and conventions.

You will learn to earn these points step by step in the **SCR Mastery** and **ECR Mastery** sections, which include real-style prompts and sample responses scored at high, middle, and low levels.

Tracking Your Own Scores

After each practice test, use the **Score-Tracking Sheets** and **Strand Diagnostic Charts** at the back. They turn raw points into a performance estimate and show you **which strand or skill to study next.** Watching your scores climb from Test 1 to Test 4 is the clearest sign your work is paying off.

Item-Type Field Guide

How Online Questions Look on Paper

STAAR is taken on a computer, so some questions ask you to do more than bubble a letter. This guide shows **every question type you may meet**, how we present it on paper in this book, and a quick example. The skill tested is always a Grade 8 TEKS skill — only the way you record your answer changes.

> **Key Idea:** On screen you *click, type, or drag.* On paper you *circle, write, or draw an arrow.* The thinking is exactly the same.

1. Multiple Choice

Online: Click one choice (A–D). **On paper:** Circle the letter of the one best answer.

Example (Math). Which number is the greatest? A. 0.6 B. ⅗ C. 0.58 D. ½ *(Answer: A)*

2. Multiselect

Online: Check every correct box. **On paper:** Circle **all** correct letters; the question says how many.

Example (Math). Select the **two** numbers that are irrational. A. √49 B. √10 C. π D. 0.25 *(Answers: B and C)*

3. Inline Choice (Drop-Down)

Online: Click a drop-down inside the sentence. **On paper:** Each drop-down is a box of choices; circle the correct one.

Example (Revising & Editing). The science club meets after school, and the members [is / **are** / was] planning a fundraiser. *(Answer: are)*

4. Text Entry / Griddable

Online: Type a number or short answer; for numbers you may fill a grid. **On paper:** Write in the box, then bubble the grid when provided.

Example (Math). A line passes through (0, 3) with slope 2. What is y when $x = 4$? [ ] *(Answer: 11)*

5. Drag-and-Drop

Online: Drag words/numbers from a bank into blanks or an order. **On paper:** A bank is printed; write each item on the correct line or draw an arrow.

Example (Math). Order from **least to greatest.** Bank: 0.4, ⅓, 0.37, ⅖ Least __ __ __ __ Greatest *(Answer: ⅓, 0.37, 0.4, ⅖)*

6. Equation Editor

Online: Build an expression/equation with an on-screen keypad. **On paper:** Write it neatly on the line.

Example (Math). Write an equation in form $y = mx + b$ for slope 3, y-intercept −5. _____ *(Answer: $y = 3x - 5$)*

7. Graphing

Online: Plot points, lines, or bars on a grid. **On paper:** Use the printed coordinate grid and a pencil.

Example (Math). On the grid provided, plot (−2, 4) and draw the line $y = x + 1$.

8. Hot Text

Online: Click a sentence or phrase in the passage. **On paper:** Underline, or write the number of, the sentence that answers the question.

Example (Reading). Underline the sentence that **best** supports the idea that the narrator felt nervous.

9. Written Responses (SCR and ECR)

Online: Type in a text box. **On paper:** Write in the lined space provided.

- **Short Constructed Response (SCR)** — a few sentences with text evidence (2 points).
- **Extended Constructed Response (ECR)** — a full multi-paragraph essay (10 points).

You will practice these in depth in the **SCR Mastery** and **ECR Mastery** sections.

> **Strategy:** When a question says "Select TWO" or "least to greatest," underline those words before you answer. Many points are lost not from wrong math or reading, but from missing the directions.

Your Study Calendars

Pick the plan that fits the time you have. Each uses the same four stages — **Learn, Practice, Simulate, Diagnose** — and spreads the eight full-length tests so you never cram. Check off each day as you go.

> **Key Idea:** Short, steady sessions beat one long rush. Aim for 30–45 focused minutes per study day.

The 4-Week Sprint (test is coming soon)

Week	Focus	Do This
1	Math — Strands 1 & 2	Read lessons; warm-up drills; take **Math Test 1**; review every explanation
2	Math — Strands 3 & 4	Lessons; drills; take **Math Tests 2 & 3**; fill diagnostic charts; redo weak skills
3	RLA — Reading + SCR	Genre guide & reading lessons; **SCR Mastery**; take **RLA Tests 1 & 2**
4	RLA — Writing + ECR, then finish	Revising/editing refresher; **ECR Mastery**; take **Math Test 4**, **RLA Tests 3 & 4**; final review

The 8-Week Plan (more time, deeper mastery)

Week	Focus	Do This
1	Math Strand 1 — Numerical	Lesson + drills; begin a daily routine
2	Math Strand 2 — Computations & Algebraic	Lesson + drills; **Math Test 1** + review
3	Math Strand 3 — Geometry & Measurement	Lesson + drills; **Math Test 2** + diagnostic chart
4	Math Strand 4 — Data & Financial Literacy	Lesson + drills; **Math Test 3**; reteach weak skills
5	RLA Reading — genres & strategies	Genre guide + reading lessons; hot-text practice

6	RLA Short Constructed Response	**SCR Mastery**; **RLA Test 1** + review
7	RLA Revising & Editing	Conventions refresher; **RLA Test 2**; **Math Test 4**
8	RLA Extended Constructed Response + final push	**ECR Mastery**; **RLA Tests 3 & 4**; final diagnostic review

Watch Out: Do not skip the review step. The biggest score gains come from understanding **why** you missed a question, then practicing that exact skill again.

Test-Day Game Plan

You have practiced the skills and the question types. Test day is simply one more practice test — this time for real. Follow this plan to stay calm and finish strong.

The Night Before

- Pack what you need and go to bed on time. Rest helps your brain more than last-minute cramming.
- Do a light review of your **diagnostic charts** only — no new material.

The Morning Of

- Eat a good breakfast. Arrive early so you feel settled, not rushed.
- Take a few slow breaths before you begin. You are ready.

During the Test — Your Pacing Plan

1. **Read the directions of each question carefully.** Watch for *select two*, *least to greatest*, and *best*.
2. **Answer the questions you know first.** Flag harder ones and return — don't let one question eat your time.
3. **Use your tools.** On Math, use the **calculator** and **reference sheet** and show work on scratch paper. On Reading, use the online **dictionary** when a word blocks understanding.
4. **Eliminate wrong answers.** Crossing out two choices doubles your odds on a tricky question.
5. **Never leave a question blank.** There is no penalty for guessing. Make your best choice and flag it to revisit.

Budgeting Time for Written Responses (RLA)

- Keep **SCR** answers short and direct — make your point, give **text evidence**, done.
- Save time for the **ECR** essay (10 points). A quick plan — thesis, two or three supporting points with evidence, a conclusion — keeps you organized. A partial, organized essay still earns points; a blank one earns none.

Before You Submit

- Return to every flagged question; make sure each has an answer.

- Check that multiselect questions have the **right number** of choices marked.
- Reread written responses once for clear sentences, correct spelling, and punctuation.

Key Idea: Finish every question with *some* answer, give the written responses real effort, and use your full time. Steady and complete beats fast and blank.

You've got this. Now turn the page and start building your skills.

Part 1 · Mathematics

How Grade 8 Math Is Built

The Grade 8 Mathematics test has **40 questions worth 48 points.** Every question is drawn from one of **four strands** (the state calls them reporting categories). Knowing the strands tells you exactly what to study — and this book teaches **every skill in every strand.**

Strand	What It Covers	On the Test
1 · Numerical Representations & Relationships	Real numbers, irrational numbers, scientific notation, ordering numbers	~4 questions
2 · Computations & Algebraic Relationships	Slope, linear equations, functions, one-variable equations	~15 questions (largest)
3 · Geometry & Measurement	Dilations, transformations, volume, surface area, Pythagorean theorem, angles	~13 questions
4 · Data Analysis & Personal Financial Literacy	Scatterplots, trend lines, mean absolute deviation, interest, saving	~8 questions

Readiness vs. Supporting Standards

- ★ **Readiness standard** — tested heavily every year; carries most of the points. Master these first.
- ○ **Supporting standard** — tested less often, but still fair game.

The Process Standards (8.1 A–G)

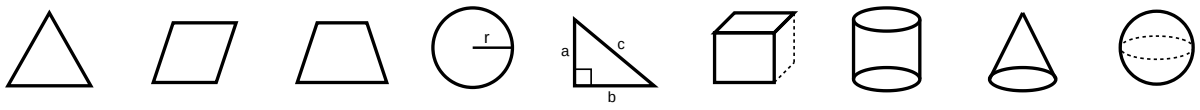
STAAR also weaves in **mathematical process standards**: applying math to real situations, choosing a strategy, using tools, and explaining your reasoning. You practice these inside every lesson and test — they are *how* you do the math, not a separate topic.

> **Key Idea:** Read each lesson, do the warm-up drills, then prove it on the practice tests. Use the **reference sheet** on the next page the whole way through — it is yours on the real test, too.

Grade 8 Mathematics Reference Sheet

On the real STAAR test, these formulas are provided for you. Learn where each one is so you can find it fast. (This replicates the key formulas from the STAAR Grade 8 Mathematics Reference Materials.)

Shapes at a glance



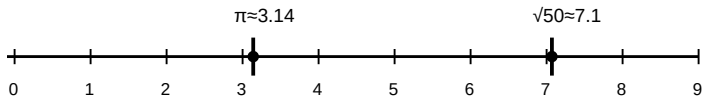
Triangle · Parallelogram · Trapezoid · Circle · Right triangle · Prism · Cylinder · Cone · Sphere

Perimeter & Circumference	Area
Circle: $\mathbf{C = 2\pi r}$ or $\mathbf{C = \pi d}$	Triangle: $\mathbf{A = \frac{1}{2}bh}$ Rectangle / Parallelogram: $\mathbf{A = bh}$ Trapezoid: $\mathbf{A = \frac{1}{2}(b_1 + b_2)h}$ Circle: $\mathbf{A = \pi r^2}$
Surface Area (B = base area, P = base perimeter)	**Volume** (B = base area)
Prism — Lateral: $\mathbf{S = Ph}$ · Total: $\mathbf{S = Ph + 2B}$ Cylinder — Lateral: $\mathbf{S = 2\pi rh}$ · Total: $\mathbf{S = 2\pi rh + 2\pi r^2}$	Prism / Cylinder: $\mathbf{V = Bh}$ Cone: $\mathbf{V = \frac{1}{3}Bh}$ Sphere: $\mathbf{V = \frac{4}{3}\pi r^3}$
Algebra & Graphing	**Personal Financial Literacy**
Slope: $\mathbf{m = (y_2 - y_1) / (x_2 - x_1)}$ Slope-intercept form: $\mathbf{y = mx + b}$ Direct variation: $\mathbf{y = kx}$ Pythagorean theorem: $\mathbf{a^2 + b^2 = c^2}$	Simple interest: $\mathbf{I = Prt}$ Compound interest: $\mathbf{A = P(1 + r/n)^{nt}}$
Useful Constants & Conversions	
$\mathbf{\pi \approx 3.14}$ (or $22/7$) · Metric: 1 m = 100 cm = 1000 mm; 1 km = 1000 m · Customary: 1 ft = 12 in; 1 yd = 3 ft; 1 mi = 5,280 ft	

Strategy: You do **not** need to memorize these — but you must know they exist and how to use them. Every worked example that uses a formula shows you which one and why.

Strand 1 · Numerical Representations and Relationships

Placing irrational numbers on a number line (8.2B)



What this strand tests: how to represent, compare, and order real numbers — including irrational numbers and scientific notation.

Standards covered:

- ○ **8.2A** — Describe relationships between sets of real numbers using a visual representation.
- ○ **8.2B** — Approximate an irrational number (including π and square roots of numbers less than 225) and locate it on a number line.
- ○ **8.2C** — Convert between standard decimal notation and scientific notation.
- ★ **8.2D** — Order a set of real numbers from mathematical and real-world contexts. *(Readiness)*

8.2A · The Real Number System ○

Every number you use at Grade 8 is a **real number.** Real numbers divide into groups nested inside one another:

- **Natural:** 1, 2, 3, … **Whole:** 0, 1, 2, 3, … **Integers:** … −2, −1, 0, 1, 2, …
- **Rational:** any number written as a fraction a/b — integers, terminating decimals (0.75), repeating decimals (0.333…).
- **Irrational:** cannot be written as a simple fraction; decimals never end or repeat — such as **π** and **√2**.

Key Idea: The groups are *nested* — every natural number is also whole, an integer, and rational. But rational and irrational never overlap.

Worked example. Classify √16 and √20. √16 = 4 → natural, whole, integer, **rational.** √20 ≈ 4.472… → **irrational.**

Watch Out: A square root is irrational only when the number is **not** a perfect square. √25 = 5 is rational; √26 is irrational.

Try It. Every integer is a rational number — true or false? *(True: any integer $n = n/1$.)*

8.2B · Approximating Irrational Numbers ○

To estimate a square root, find the two **perfect squares** it falls between.

Worked example. Approximate √50. $49 = 7^2$ and $64 = 8^2$, so √50 is between **7 and 8**; just above 49, so √50 ≈ **7.1** — mark it just right of 7 on a number line. **π ≈ 3.14**, between 3 and 4.

Watch Out: "Less than 225" means the number under the root is below 225 — roots up to about √224 ≈ 14.97.

Try It. Between which two whole numbers is √30? *(5 and 6, since 25 < 30 < 36.)*

8.2C · Scientific Notation ○

Scientific notation writes a number as $\mathbf{a \times 10^n}$, where $\mathbf{1 \le |a| < 10}$ and n is an integer. Large numbers → positive exponent; small numbers (less than 1) → negative exponent.

- 4,500,000 = $\mathbf{4.5 \times 10^6}$ · 0.00032 = $\mathbf{3.2 \times 10^{-4}}$ · 7.1×10^5 = **710,000**

Watch Out: The factor *a* must be at least 1 and less than 10. Writing 45×10^5 is wrong — fix to 4.5×10^6.

Try It. Write 0.0067 in scientific notation. *(6.7×10^{-3}.)*

8.2D · Ordering Real Numbers ★ (Readiness)

Numbers often appear in different forms at once — fractions, decimals, percents, roots, scientific notation. **Convert them all to decimals, then compare.**

Worked example. Order from least to greatest: ⅗, 0.58, √0.36, 62%.
⅗ = 0.60 · 0.58 = 0.58 · √0.36 = 0.60 · 62% = 0.62 → **0.58, then ⅗ = √0.36, then 62%.**

> **Key Idea:** One common form — usually the decimal — lets you compare very different-looking numbers fairly. A number line helps you *see* the order.

Watch Out: On a negative number line, the number farther **left** is **smaller**. −0.8 < −0.3.

Try It. Order greatest to least: 0.7, ⅔, 71%. *(71% = 0.71, 0.7, ⅔ ≈ 0.667.)*

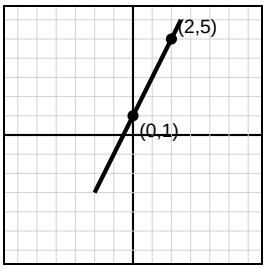
Strand 1 Checklist

- ☐ 8.2A Classify real numbers and describe how the sets relate
- ☐ 8.2B Approximate irrational numbers and place them on a number line
- ☐ 8.2C Convert between standard and scientific notation
- ☐ 8.2D Order a mixed set of real numbers

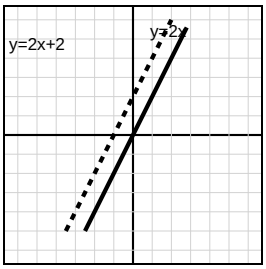
Ready to practice? Warm-up drills for Strand 1 come after all four strand lessons.

Strand 2 · Computations and Algebraic Relationships

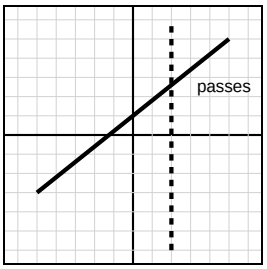
Key Figures for This Strand



Slope from a graph (8.4C)



Proportional vs. not (8.5A/B)



Vertical line test (8.5G)

This is the **largest strand** on the test (about 15 of 40 questions). It is all about **lines, relationships, and equations.** Master it and you master most of the Math test.

Standards covered:

- ○ **8.4A** — Similar right triangles show slope is the same between any two points on a line.
- ★ **8.4B** — Graph proportional relationships; interpret the unit rate as the slope.
- ★ **8.4C** — Use a table or graph to find the rate of change (slope) and the y-intercept.
- ○ **8.5A** — Linear proportional situations as y = kx. ○ **8.5B** — Non-proportional as y = mx + b (b ≠ 0).
- ○ **8.5E** — Direct variation. ○ **8.5F** — Distinguish proportional from non-proportional. ○ **8.5H** — Examples of each.
- ★ **8.5G** — Identify functions from ordered pairs, tables, mappings, graphs.
- ★ **8.5I** — Write an equation y = mx + b to model a linear relationship.
- ○ **8.8A** — Write one-variable equations/inequalities with variables on both sides. ○ **8.8B** — Write a real-world problem for a given equation. ★ **8.8C** — Model and solve them.
- ○ **8.9A** — Identify a system's solution from the intersection of two graphs.

Part A · Slope and Rate of Change (8.4A ○, 8.4B ★, 8.4C ★)

Slope measures how steep a line is — how much *y* changes for each change in *x*:

$$\textbf{slope = m = (change in y) ÷ (change in x) = } (y_2 - y_1) / (x_2 - x_1)$$

- **8.4A — Why slope is constant.** Right triangles drawn to any two points on the same line are **similar**, so rise ÷ run is the **same everywhere** — that ratio is the slope.

- **8.4B — Proportional graphs.** A proportional relationship passes through the **origin (0, 0)**; equation **y = kx**; the **unit rate is the slope k.**
- **8.4C — From a table or graph.** Slope = (change in y) ÷ (change in x); the **y-intercept** is the y-value when **x = 0.**

Worked example (8.4C). Table: (0, 2), (1, 5), (2, 8). Slope = (5 − 2) ÷ (1 − 0) = **3**; y-intercept = **2**; equation y = 3x + 2.

Watch Out: Rise over run — count the **vertical** change first, then horizontal.

Try It. A line through (0, 0) and (4, 10) — slope? *(10 ÷ 4 = 2.5.)*

Part B · Proportional vs. Non-Proportional (8.5A/B/E/F/H ○)

	Proportional	Non-Proportional
Equation	y = kx	y = mx + b (b ≠ 0)
Graph	Line **through the origin**	Line **not** through the origin
Table	Constant ratio $y/x = k$	y/x is **not** constant

- **8.5E — Direct variation** is a proportional relationship y = kx. If y = 15 when x = 3, then k = 5, so y = 5x.
- **8.5F / 8.5H — Telling them apart.** Does the line pass through (0, 0)? Yes → proportional. A starting amount (y-intercept ≠ 0) → non-proportional.

Worked example (8.5B). A gym charges a $20 sign-up fee plus $5 per visit: **y = 5x + 20.** The $20 fee (b ≠ 0) makes it **non-proportional.**

Watch Out: A per-item rate with no starting fee is proportional; add any flat fee and it becomes non-proportional.

Try It. Is y = 7x proportional? *(Yes — passes through the origin, b = 0.)*

Part C · Functions (8.5G ★)

A **function** pairs **each input (x) with exactly one output (y).** In tables/mappings, an x-value may not repeat with two different y-values. On graphs, use the **vertical line test** — if any vertical line crosses the graph more than once, it is **not** a function.

Worked example. Is {(1, 3), (2, 5), (1, 7)} a function? **No** — input 1 pairs with both 3 and 7.

Try It. Does a non-vertical straight line pass the vertical line test? *(Yes — it is a function.)*

Part D · Writing Linear Equations (8.5I ★)

To write **y = mx + b**, find the **slope m** and the **y-intercept b.**

Worked example. A plant is 4 cm tall and grows 2 cm each week. Slope m = **2**; y-intercept b = **4**; equation **y = 2x + 4.**

Strategy: "Starting amount" is almost always **b**; "rate per unit" is almost always **m.**

Try It. A taxi charges $3 to start plus $2 per mile. Equation for cost y after x miles? *(y = 2x + 3.)*

Part E · One-Variable Equations and Inequalities (8.8A/B ○, 8.8C ★)

When the variable is on **both sides,** collect variable terms on one side and numbers on the other.

Worked example (8.8C). Solve 3x + 5 = x − 7. Subtract x → 2x + 5 = −7 Subtract 5 → 2x = −12 Divide by 2 → **x = −6.**

8.8A write such equations from a situation; **8.8B** do the reverse — describe a real situation the equation could model.

Watch Out: Whatever you do to one side, do to the **other** side — keep it balanced.

Try It. Solve 4x − 3 = 2x + 9. *(2x = 12, x = 6.)*

Part F · Systems of Two Linear Equations (8.9A ○)

Two lines graphed together **intersect** at the point that **satisfies both equations.** That point **(x, y)** is the **solution** of the system.

Worked example. y = x + 1 and y = −x + 5 cross at **(2, 3).** Check: 3 = 2 + 1 ✓ and 3 = −2 + 5 ✓.

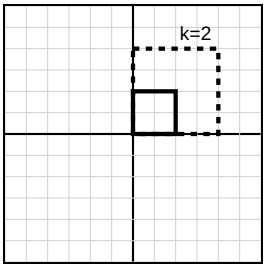
Try It. If two lines cross at (−1, 4), what is the solution? *(x = −1, y = 4.)*

Strand 2 Checklist

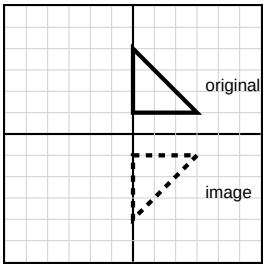
- ☐ 8.4A–C Find and interpret slope, unit rate, and y-intercept
- ☐ 8.5A/B/E/F/H Tell proportional from non-proportional; use y = kx and y = mx + b
- ☐ 8.5G Identify functions (including the vertical line test)
- ☐ 8.5I Write an equation for a linear relationship
- ☐ 8.8A–C Write and solve one-variable equations with variables on both sides
- ☐ 8.9A Identify a system's solution from the intersection of two graphs

Strand 3 · Geometry and Measurement

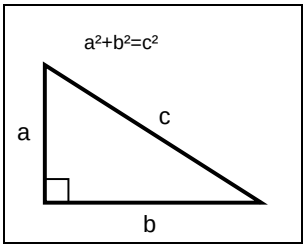
Key Figures for This Strand



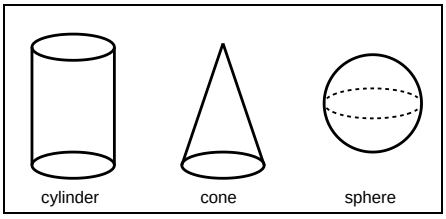
Dilation (8.3C, 8.10D)



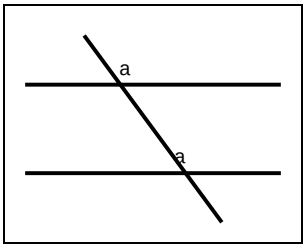
Reflection (8.10C)



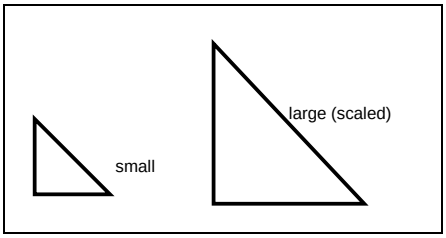
Pythagorean (8.7C)



Volume solids (8.7A)



Transversal angles (8.8D)



Similar figures (8.3A)

This strand (about 13 of 40 questions) covers **dilations, transformations, angles, volume, surface area, and the Pythagorean theorem.** Keep the **reference sheet** open — most of these use a formula.

Standards covered:

- ○ **8.3A** — Ratio of corresponding sides of similar shapes is proportional. ○ **8.3B** — Compare a shape and its dilation on a plane. ★ **8.3C** — Algebraic rule for a scale factor (origin center).
- ○ **8.6A** — Volume formula V = Bh of a cylinder. ○ **8.6C** — Explain the Pythagorean theorem with models.
- ★ **8.7A** — Volume of cylinders, cones, spheres. ★ **8.7B** — Surface area of prisms and cylinders. ★ **8.7C** — Pythagorean theorem and its converse. ○ **8.7D** — Distance between two points.
- ○ **8.8D** — Angle sums, exterior angles, transversal angles, AA similarity.
- ○ **8.10A** — Orientation/congruence of transformations. ○ **8.10B** — Which preserve congruence. ★ **8.10C** — Rules for translations, reflections, rotations (90/180/270/360). ○ **8.10D** — Effect of a dilation on length and area.

Part A · Dilations and Similarity (8.3A ○, 8.3B ○, 8.3C ★, 8.10D ○)

A **dilation** resizes a figure by a **scale factor k** from a center. With the **origin** as center: **(x, y) → (kx, ky).**

- **8.3A / 8.3B — Similar shapes.** A figure and its dilation are **similar**: corresponding sides are proportional (×k) and angles stay the **same.**

- **8.10D — Effect on measurements.** Lengths multiply by **k**; **area** multiplies by $\mathbf{k^2}$.

Worked example (8.3C). Vertex (2, 3) dilated by factor 3 about the origin → (3·2, 3·3) = **(6, 9).**

Watch Out: k > 1 enlarges; 0 < k < 1 shrinks. Area changes by the **square** of k — factor 2 makes area **4×**.

Try It. A square of side 5 is dilated by k = 2 — new side? *(10.)*

Part B · Transformations on the Coordinate Plane (8.10A/B ○, 8.10C ★)

Transformation	Algebraic rule	Changes size?
Translation (slide)	(x, y) → (x + a, y + b)	No — congruent
Reflection over x-axis	(x, y) → (x, −y)	No — congruent
Reflection over y-axis	(x, y) → (−x, y)	No — congruent
Rotation 180° about origin	(x, y) → (−x, −y)	No — congruent
Rotation 90° counterclockwise	(x, y) → (−y, x)	No — congruent
Dilation (factor k)	(x, y) → (kx, ky)	**Yes** — similar, not congruent

8.10A / 8.10B — Congruence. Translations, reflections, and rotations are **rigid** — the image is **congruent.** A dilation (k ≠ 1) changes size, so it is **not** congruent, only **similar.**

Worked example (8.10C). Reflect (4, −2) over the x-axis → **(4, 2).**

Watch Out: Reflecting over the **x-axis** flips the **y**-sign; over the **y-axis** flips the **x**-sign.

Try It. Rotate (3, 5) by 180° about the origin. *((−3, −5).)*

Part C · Angle Relationships (8.8D ○)

- **Triangle angle sum:** interior angles add to **180°.**
- **Exterior angle** = sum of the two remote interior angles.
- **Parallel lines + transversal:** corresponding angles equal; alternate interior angles equal.
- **Angle-Angle (AA) similarity:** two equal angles → triangles are **similar.**

Worked example. Angles 50° and 60° → third = 180 − 50 − 60 = **70°.**

Try It. Angles 90° and 35° — find the third. *(55°.)*

Part D · Volume (8.6A ○, 8.7A ★)

- **Cylinder:** V = Bh = $\pi r^2 h$. **Cone:** V = $\frac{1}{3}\pi r^2 h$ (one-third of the cylinder). **Sphere:** V = $\frac{4}{3}\pi r^3$.

Worked example (8.7A). Cylinder radius 3, height 10: V = $\pi(3^2)(10)$ = **$90\pi \approx 282.6$** cubic units.

Watch Out: The cone is ⅓ of the matching cylinder — don't forget the one-third.

Try It. Cone radius 3, height 10 — volume? *(⅓·90π = 30π ≈ 94.2.)*

Part E · Surface Area (8.7B ★)

- **Prism — total:** S = Ph + 2B. **Cylinder — total:** S = 2πrh + 2πr^2 (side plus two circles).

Worked example (8.7B). Cylinder radius 2, height 5: S = 2π(2)(5) + 2π(2^2) = 20π + 8π = **28π ≈ 87.9** square units.

Strategy: "Lateral" = the **side** only; "total" **adds the bases.** Read which one is asked.

Try It. Lateral surface area of that cylinder? *(2πrh = 20π ≈ 62.8.)*

Part F · The Pythagorean Theorem (8.6C ○, 8.7C ★, 8.7D ○)

For a **right triangle** with legs a and b and hypotenuse c: **$a^2 + b^2 = c^2$.**

- **8.7C** — Find a missing side, or use the **converse** to test whether a triangle is right.
- **8.7D** — Distance between two points: horizontal and vertical gaps are the legs; the distance is the hypotenuse.

Worked example (8.7C). Legs 6 and 8: c^2 = 36 + 64 = 100 → c = **10.** **(8.7D)** Distance (1, 2) to (4, 6): legs 3 and 4 → √(9 + 16) = **5.**

Watch Out: The hypotenuse **c** is across from the right angle and is the **longest** side.

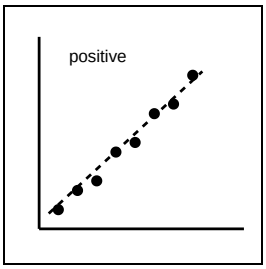
Try It. Is 5, 12, 13 a right triangle? *(Yes — 25 + 144 = 169 = 13^2.)*

Strand 3 Checklist

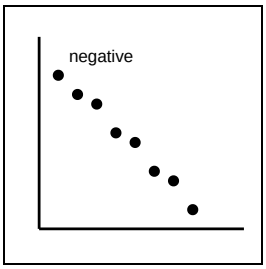
- ☐ 8.3A–C, 8.10D Dilate figures; use scale factor; length ×k and area ×k^2
- ☐ 8.10A–C Apply translation, reflection, rotation rules; know what stays congruent
- ☐ 8.8D Use angle sums, exterior angles, transversals, AA similarity
- ☐ 8.6A, 8.7A Volume of cylinders, cones, spheres
- ☐ 8.7B Lateral and total surface area of prisms and cylinders
- ☐ 8.6C, 8.7C, 8.7D Pythagorean theorem, converse, and distance

Strand 4 · Data Analysis and Personal Financial Literacy

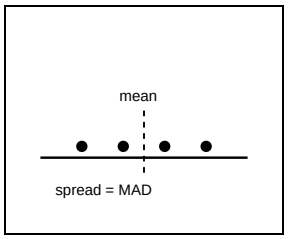
Key Figures for This Strand



Positive trend (8.5D)



Negative association (8.11A)



Mean absolute deviation (8.11B)

This strand (about 8 of 40 questions) has two halves: **reading data** (scatterplots, trend lines, spread) and **personal finance** (interest, saving, the cost of borrowing).

Standards covered:

- ○ **8.5C** — Contrast bivariate data that suggest a linear relationship with data that do not. ★ **8.5D** — Use a trend line to make predictions.
- ○ **8.11A** — Construct a scatterplot and describe the association. ○ **8.11B** — Mean absolute deviation (≤10 points).
- ○ **8.12A** — How rate and loan length affect the cost of credit. ○ **8.12C** — How regular saving grows. ★ **8.12D** — Simple vs. compound interest. ○ **8.12G** — Estimate college cost and a savings plan.

Part A · Scatterplots and Association (8.5C ○, 8.11A ○)

A **scatterplot** graphs paired (bivariate) data. The pattern shows the **association**: **positive linear** (points rise), **negative linear** (points fall), **non-linear** (a curve), or **no association** (scattered).

Worked example. Study time up, scores up → points rise right → **positive linear.** **Try It.** TV hours up, grades down → *negative linear.*

Part B · Trend Lines and Predictions (8.5D ★)

A **trend line** (line of best fit) runs through the middle of the data; use it to **predict** values you did not measure.

Worked example. Trend line y = 2x + 3 (x = weeks). Height at week 10: y = 2(10) + 3 = **23 cm.** **Watch Out:** a prediction is an **estimate**, not exact.

Part C · Mean Absolute Deviation (MAD) (8.11B ○)

MAD = the **average distance of each value from the mean.** Steps: (1) find the mean, (2) find each distance from the mean, (3) average those distances.

Worked example. Data 2, 4, 6, 8. Mean = 5. Distances 3, 1, 1, 3. MAD = 8 ÷ 4 = **2.** **Try It.** MAD of 10, 10, 20, 20? *(Mean 15; MAD = 5.)*

Part D · Personal Financial Literacy (8.12A/C/G ○, 8.12D ★)

Simple interest: $I = Prt$. **Compound interest:** $A = P(1 + r/n)^{nt}$ — interest earns interest, so it grows faster.

Worked example (8.12D). $500 at 4% for 3 years, **simple**: I = 500 × 0.04 × 3 = **$60.** Compound would earn slightly more.

- **8.12A — Cost of credit.** A higher rate or a longer loan means you pay **more** interest overall.
- **8.12C — Saving regularly.** Small, steady amounts add up and grow — especially with compound interest and time.
- **8.12G — College savings.** Estimate total cost, subtract expected family contribution, divide the rest across the months you have.

Try It. Simple interest on $1,000 at 5% for 2 years? *(1000 × 0.05 × 2 = $100.)*

Strand 4 Checklist

- ☐ 8.5C, 8.11A Read scatterplots and name the association
- ☐ 8.5D Use a trend line to predict · ☐ 8.11B Calculate mean absolute deviation
- ☐ 8.12A/C/D/G Simple vs. compound interest, cost of credit, saving and college planning

Process Standards in Action (8.1 A–G)

The **mathematical process standards** are not a separate topic — they are *how* you do every problem. STAAR builds them into questions across all four strands.

Standard	What it asks you to do	How you use it
8.1A	Apply math to everyday life, society, work	Solve real-world problems (cost, distance, savings)
8.1B	Use a problem-solving model	Analyze, plan, solve, check reasonableness
8.1C	Choose the right tools and techniques	Calculator, mental math, estimation, reference sheet
8.1D	Communicate with multiple representations	Move between symbols, tables, graphs, words
8.1E	Create and use representations	Draw a diagram, make a table, sketch a graph
8.1F	Analyze relationships to connect ideas	See slope, tables, equations as the same line
8.1G	Justify reasoning with precise language	Explain *why* an answer is correct

Key Idea: Strong test-takers do more than "get the answer." They **check that it makes sense,** choose efficient tools, and can explain their thinking. Practice that on every item.

Strategy: After each answer ask: *Does this make sense?* A volume should be positive, a probability between 0 and 1, a slope matching the line's direction. Catching an unreasonable answer saves easy points.

Warm-Up Skill Drills

Short, mixed drills to warm up before the full-length tests. Do a set, then check the answer key. Each item names the TEKS it practices.

Strand 1 — Numerical Representations and Relationships

1. **(8.2D)** Order from least to greatest: 0.45, ⅖, √0.2, 43%.
2. **(8.2B)** Between which two whole numbers is √72?
3. **(8.2C)** Write 0.00056 in scientific notation.
4. **(8.2A)** Is √81 rational or irrational?
5. **(8.2C)** Write 3.2×10^5 in standard form.

Strand 2 — Computations and Algebraic Relationships

6. **(8.4C)** Find the slope of the line through (2, 3) and (6, 11).
7. **(8.5I)** A tank holds 50 L and drains 4 L per minute. Write an equation for amount y after x minutes.
8. **(8.5G)** Is {(2, 4), (3, 9), (2, 5)} a function?
9. **(8.8C)** Solve $5x - 2 = 3x + 8$.
10. **(8.5F)** Is $y = 6x$ proportional or non-proportional?

Strand 3 — Geometry and Measurement

11. **(8.7C)** A right triangle has legs 9 and 12. Find the hypotenuse.
12. **(8.7A)** Volume of a cylinder with radius 5, height 4 ($\pi \approx 3.14$).
13. **(8.10C)** Reflect (−3, 7) over the y-axis.
14. **(8.3C)** Dilate (4, −6) by scale factor ½ about the origin.
15. **(8.8D)** Two angles of a triangle are 42° and 73°. Find the third.

Strand 4 — Data Analysis and Personal Financial Literacy

16. **(8.12D)** Simple interest on $500 at 4% for 3 years.
17. **(8.11B)** Mean absolute deviation of 2, 4, 6, 8.
18. **(8.5D)** Trend line $y = 2x + 3$. Predict y when x = 10.
19. **(8.11A)** Points rise from lower-left to upper-right — name the association.
20. **(8.12A)** True or false: a longer loan at the same rate usually costs more in total interest.

Answer Key — Warm-Up Drills

1. ⅖ (0.40), 43% (0.43), √0.2 (≈0.447), 0.45 · 2. 8 and 9 · 3. 5.6×10^{-4} · 4. Rational (= 9) · 5. 320,000 · 6. m = 2 · 7. y = −4x + 50 · 8. No (input 2 → two outputs) · 9. x = 5 · 10. Proportional · 11. c = 15 · 12. 314 cubic units · 13. (3, 7) · 14. (2, −3) · 15. 65° · 16. $60 · 17. 2 · 18. 23 · 19. Positive linear · 20. True

Mathematics · Practice Test 1

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Which list is ordered from **least to greatest**?

A. ⅔, 68%, 0.7, √0.5 B. √0.5, 0.7, 68%, ⅔ C. 68%, ⅔, √0.5, 0.7 D. 0.7, √0.5, ⅔, 68%

2. Between which two whole numbers does √40 lie?

A. 5 and 6 B. 6 and 7 C. 19 and 21 D. 20 and 21

3. *(Griddable)* Write 68,000 in scientific notation. ☐

4. Which number is **irrational**? A. √36 B. ⅘ C. π D. 0.25

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (1, 2) and (4, 11)? A. 2 B. 3 C. 6 D. 9

6. A proportional relationship passes through the origin and the point (3, 12). What is the unit rate? A. 3 B. 4 C. 9 D. 15

7. Why is the slope of a straight line the same between any two points?

A. The line is horizontal B. The right triangles formed are **similar**, so rise÷run is constant C. All lines have slope 1 D. The points are always 1 unit apart

8. *(Equation editor)* A plant is 6 cm tall and grows 3 cm per week. Write an equation for its height y after x weeks. ☐

9. Which set of ordered pairs represents a **function**?

A. {(1, 2), (1, 3), (2, 4)} B. {(2, 5), (3, 5), (4, 5)} C. {(0, 1), (0, 2), (0, 3)} D. {(5, 1), (5, 2), (6, 3)}

10. Which equation represents a **proportional** relationship? A. $y = 4x$ B. $y = 4x + 2$ C. $y = x - 4$ D. $y = 2x - 1$

11. A gym charges a $15 membership fee plus $6 per visit. Which equation gives the cost y for x visits? A. $y = 15x + 6$ B. $y = 6x + 15$ C. $y = 21x$ D. $y = 6x - 15$

12. *(Griddable)* For a direct variation, $y = 20$ when $x = 5$. What is y when $x = 9$? ☐

13. Which table shows a **non-proportional** relationship?

A. (1, 3)(2, 6)(3, 9) B. (1, 5)(2, 10)(3, 15) C. (1, 4)(2, 7)(3, 10) D. (2, 8)(3, 12)(4, 16)

14. Which situation is **proportional**?

A. A taxi charges $3 plus $2 per mile B. A pool has 10 L and drains 2 L per minute C. Apples cost $2 per pound D. A phone plan is $20 plus $0.10 per text

15. *(Griddable)* Solve for x: $6x + 4 = 2x + 20$. ☐

16. "Three less than twice a number equals the number plus five." Which equation models this? A. $2x - 3 = x + 5$ B. $3 - 2x = x + 5$ C. $2x + 3 = x - 5$ D. $3x - 2 = x + 5$

17. Which situation can be modeled by $5x + 10 = 3x + 30$?

A. Two savers start with $10 and $30 and save $5 and $3 per week; when are the totals equal? B. A number times 5 C. The area of a rectangle D. The sum of 10 and 30

18. Two lines are graphed and cross at (2, 5). What is the solution of the system? A. $x = 5, y = 2$ B. $x = 2, y = 5$ C. $x = 7$ D. No solution

19. *(Select TWO)* Which statements about the line $y = 2x - 3$ are true? A. The slope is 2 B. The slope is −3 C. The y-intercept is −3 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 4 and height 7. What is its volume? ($\pi \approx 3.14$) A. 87.92 B. 175.84 C. 351.68 D. 703.36

21. A cone has radius 6 and height 10. What is its volume? ($\pi \approx 3.14$) A. 125.6 B. 376.8 C. 1130.4 D. 3768

22. A cylinder has radius 3 and height 8. What is its **total** surface area? ($\pi \approx 3.14$) A. 150.72 B. 169.56 C. 207.24 D. 263.76

23. *(Griddable)* A right triangle has legs of 7 and 24. What is the length of the hypotenuse? ☐

24. What is the distance between (−1, −3) and (3, 0)? A. 5 B. 7 C. 12 D. 25

25. The point (−2, 5) is dilated by a scale factor of 3 about the origin. What is the image? A. (−6, 15) B. (6, 15) C. (−5, 8) D. (1, 8)

26. The point (5, −4) is reflected over the y-axis. What is the image? A. (5, 4) B. (−5, −4) C. (−5, 4) D. (4, −5)

27. The point (2, 7) is rotated 90° counterclockwise about the origin. What is the image? A. (7, 2) B. (−7, 2) C. (7, −2) D. (−2, −7)

28. Which transformation does **not** preserve congruence? A. Translation B. Reflection C. Rotation D. Dilation (scale factor 3)

29. The angles of a triangle measure 3x, 2x, and x. What is the measure of the largest angle? A. 30° B. 60° C. 90° D. 120°

30. Two similar rectangles have corresponding sides in a ratio of 2 : 5. The smaller has a length of 6. What is the matching length of the larger? A. 10 B. 12 C. 15 D. 18

31. In the cylinder volume formula $V = Bh$, what does B represent? A. $2\pi r$ B. πr^2 C. πd D. $r + h$

32. A square with side 4 is dilated by a scale factor of 3. What is the area of the image? A. 12 B. 48 C. 108 D. 144

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 3, 5, 7, 9. ☐

34. A trend line is modeled by y = 1.5x + 2. Predict *y* when x = 8. A. 11 B. 12 C. 14 D. 16

35. On a scatterplot, the points fall from the upper-left to the lower-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which description suggests a **linear** association between two variables? A. Points scattered randomly B. Points following a straight upward path C. Points forming a U-shape D. Points in a circle

37. *(Griddable)* Find the simple interest on $800 at 5% for 4 years. ☐

38. For the same loan amount and interest rate, which loan costs the **most** total interest? A. A 2-year loan B. A 3-year loan C. A 5-year loan D. They cost the same

39. How do small amounts of money invested regularly behave over time? A. They lose value B. They stay exactly the same C. They grow, especially with compound interest D. They cannot be saved

40. College costs $12,000. A family expects to contribute $4,000 and wants to save the rest over 16 months. How much must they save each month? A. $250 B. $500 C. $750 D. $1,000

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Mathematics · Practice Test 2

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Which list is ordered from **least to greatest**?

A. 0.72, ¾, √0.6, 78% B. ¾, 0.72, 78%, √0.6 C. 78%, √0.6, ¾, 0.72 D. √0.6, 0.72, ¾, 78%

2. Between which two whole numbers does √90 lie? A. 8 and 9 B. 9 and 10 C. 44 and 46 D. 89 and 91

3. *(Griddable)* Write 0.00045 in scientific notation. ☐

4. Which number is **rational**? A. √8 B. √16 C. π D. √2

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−2, 1) and (2, 9)? A. 1 B. 2 C. 4 D. 8

6. A proportional relationship passes through the origin and the point (5, 15). What is the unit rate? A. 3 B. 5 C. 10 D. 15

7. *(Equation editor)* A candle is 20 cm tall and burns down 2 cm each hour. Write an equation for its height y after x hours. ☐

8. Which relation is a **function**?

A. {(2, 1), (2, 3), (4, 5)} B. {(1, 6), (2, 6), (3, 7)} C. {(5, 2), (5, 4), (6, 8)} D. {(0, 0), (0, 1), (1, 2)}

9. Which equation represents a **non-proportional** relationship? A. y = 3x B. y = 7x C. y = 3x + 4 D. y = ½x

10. *(Griddable)* For a direct variation, y = 36 when x = 4. What is y when x = 7? ☐

11. Which table shows a **proportional** relationship?

A. (1, 2)(2, 5)(3, 8) B. (1, 4)(2, 8)(3, 12) C. (1, 3)(2, 5)(3, 7) D. (2, 7)(3, 9)(4, 11)

12. *(Griddable)* Solve for x: 7x − 5 = 3x + 19. ☐

13. "Five more than three times a number equals twice the number plus twelve." Which equation models this? A. 3x + 5 = 2x + 12 B. 5x + 3 = 12x + 2 C. 3x − 5 = 2x + 12 D. 5 + 3 = 2x + 12

14. Two lines intersect at (−3, 2). What is the solution of the system? A. x = 2, y = −3 B. x = −3, y = 2 C. x = −1 D. No solution

15. On a line, right triangles are drawn to two different pairs of points. What must be true of the triangles? A. Congruent B. **Similar**, so the slope is the same C. Equal area D. Isosceles

16. Which situation represents a **proportional** relationship?

A. A plumber charges $50 plus $40 per hour B. A runner travels 6 miles per hour at a steady pace C. A tank has 30 L and loses 5 L per hour D. A gym costs $25 to join plus $5 per class

17. Which situation can be modeled by $4x + 6 = 2x + 18$?

A. Two accounts start at $6 and $18 and grow $4 and $2 per week; when are they equal? B. The product of 4 and 6 C. A rectangle's perimeter D. 4 plus 6 plus 18

18. A taxi charges $4 to start plus $3 per mile. Which equation gives the cost y for x miles? A. $y = 4x + 3$ B. $y = 3x + 4$ C. $y = 7x$ D. $y = 3x - 4$

19. *(Select TWO)* Which statements about the line $y = -3x + 6$ are true? A. The slope is -3 B. The slope is 6 C. The y-intercept is 6 D. The y-intercept is -3

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 5 and height 6. What is its volume? ($\pi \approx 3.14$) A. 150 B. 235.5 C. 471 D. 942

21. A sphere has radius 3. What is its volume? ($\pi \approx 3.14$) A. 28.26 B. 84.78 C. 113.04 D. 339.12

22. A cylinder has radius 2 and height 10. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 150.72 C. 175.84 D. 251.2

23. *(Griddable)* A right triangle has legs of 9 and 40. What is the length of the hypotenuse? ☐

24. What is the distance between (0, 0) and (6, 8)? A. 7 B. 10 C. 14 D. 48

25. The point $(4, -3)$ is dilated by a scale factor of 2 about the origin. What is the image? A. $(6, -1)$ B. $(8, -6)$ C. $(2, -1.5)$ D. $(8, -3)$

26. The point $(-6, 2)$ is reflected over the x-axis. What is the image? A. $(6, 2)$ B. $(-6, -2)$ C. $(6, -2)$ D. $(2, -6)$

27. The point $(4, 1)$ is rotated 180° about the origin. What is the image? A. $(-4, -1)$ B. $(-1, 4)$ C. $(1, -4)$ D. $(4, -1)$

28. Which transformation **preserves** congruence (keeps the same size)? A. Dilation by factor 2 B. Dilation by factor ½ C. Translation D. Dilation by factor 3

29. Two parallel lines are cut by a transversal. One angle measures 65°. What is its corresponding angle? A. 25° B. 65° C. 115° D. 180°

30. Two similar triangles have corresponding sides in a ratio of 3 : 4. A side of the smaller is 9. What is the matching side of the larger? A. 10 B. 11 C. 12 D. 13

31. The Pythagorean theorem relates the sides of a right triangle by which equation? A. $a + b = c$ B. $a^2 + b^2 = c^2$ C. $a^2 - b^2 = c^2$ D. $2a + 2b = c$

32. A rectangle measures 5 by 3. It is dilated by a scale factor of 2. What is the area of the image? A. 30 B. 45 C. 60 D. 90

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 4, 8, 10, 14. ☐

34. A trend line is modeled by $y = 3x + 5$. Predict y when $x = 6$. A. 18 B. 20 C. 23 D. 30

35. A scatterplot shows points with no clear pattern or direction. What association is shown? A. Positive linear B. Negative linear C. No association D. Strong linear

36. Which description suggests a **non-linear** association? A. A straight upward line B. A straight downward line C. A curved, U-shaped path D. Points exactly on one line

37. *(Griddable)* Find the simple interest on $1,200 at 6% for 3 years. ☐

38. For the same loan amount and length, how does a **higher** interest rate affect the total cost? A. Lowers it B. Raises it C. No effect D. Cancels the loan

39. Which plan is likely to grow the **most** over time? A. Saving nothing B. Saving a little, once C. Saving small amounts regularly with compound interest D. Spending it all

40. College costs $16,000. A family expects to contribute $6,000 and wants to save the rest over 20 months. How much must they save each month? A. $400 B. $500 C. $600 D. $800

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Mathematics · Practice Test 3

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: 0.6, 62%, ⅝, √0.4. A. 0.6, 62%, ⅝, √0.4 B. √0.4, ⅝, 62%, 0.6 C. 62%, 0.6, ⅝, √0.4 D. ⅝, 0.6, 62%, √0.4

2. Between which two whole numbers does √115 lie? A. 9 and 10 B. 10 and 11 C. 11 and 12 D. 57 and 58

3. *(Griddable)* Write 5,200,000 in scientific notation. ☐

4. Which number is **irrational**? A. √45 B. √49 C. ¾ D. 0.5

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−1, −2) and (3, 6)? A. 2 B. 4 C. 8 D. ½

6. A proportional relationship passes through the origin and (4, 10). What is the unit rate? A. 2 B. 2.5 C. 4 D. 6

7. *(Equation editor)* A pool fills at 25 liters per minute, starting empty. Write an equation for the amount y after x minutes. ☐

8. Which relation is a **function**? A. {(3, 1), (3, 2), (4, 5)} B. {(−1, 2), (0, 2), (1, 3)} C. {(2, 4), (2, 5), (3, 6)} D. {(7, 1), (7, 2), (8, 3)}

9. Which equation represents a **non-proportional** relationship? A. y = 5x B. y = ⅓x C. y = 2x − 5 D. y = 8x

10. *(Griddable)* For a direct variation, y = 48 when x = 6. What is y when x = 5? ☐

11. Which table shows a **proportional** relationship? A. (1, 3)(2, 6)(3, 9) B. (1, 2)(2, 5)(3, 8) C. (1, 4)(2, 6)(3, 8) D. (2, 9)(3, 11)(4, 13)

12. *(Griddable)* Solve for x: 9x − 7 = 4x + 18. ☐

13. "Seven less than four times a number equals the number plus eight." Which equation models this? A. 4x − 7 = x + 8 B. 7 − 4x = x + 8 C. 4x + 7 = x − 8 D. 7x − 4 = x + 8

14. Two lines intersect at (4, −1). What is the solution of the system? A. x = −1, y = 4 B. x = 4, y = −1 C. x = 3 D. No solution

15. The slope between any two points on a line is constant because the right triangles formed are — A. congruent B. **similar** C. equilateral D. right isosceles

16. Which situation is **proportional**? A. $30 plus $10 per hour B. 4 pounds of rice per $6, at a steady rate C. 50 L draining 4 L/min D. $12 fee plus $3 each

17. Which situation can be modeled by $6x + 2 = 2x + 22$? A. Two savers start with $2 and $22, saving $6 and $2 per week; when equal? B. 6 times 2 C. A triangle's area D. 2 plus 22

18. A bike rental costs $5 plus $2 per hour. Which equation gives the cost y for x hours? A. $y = 5x + 2$ B. $y = 2x + 5$ C. $y = 7x$ D. $y = 2x - 5$

19. *(Select TWO)* Which statements about $y = 4x - 7$ are true? A. The slope is 4 B. The slope is −7 C. The y-intercept is −7 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 3 and height 10. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 141.3 C. 282.6 D. 565.2

21. A cone has radius 3 and height 7. What is its volume? ($\pi \approx 3.14$) A. 21.98 B. 65.94 C. 131.88 D. 197.82

22. A cylinder has radius 4 and height 5. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 150.72 C. 226.08 D. 351.68

23. *(Griddable)* A right triangle has legs of 8 and 15. Find the hypotenuse. ☐

24. What is the distance between (1, 2) and (5, 5)? A. 5 B. 7 C. 12 D. 25

25. The point (3, −5) is dilated by a scale factor of 2 about the origin. What is the image? A. (5, −3) B. (6, −10) C. (1.5, −2.5) D. (6, −5)

26. The point (−2, −7) is reflected over the y-axis. What is the image? A. (2, −7) B. (−2, 7) C. (2, 7) D. (7, −2)

27. The point (−3, 4) is rotated 180° about the origin. What is the image? A. (3, −4) B. (3, 4) C. (−4, 3) D. (4, −3)

28. A triangle has area 10. It is dilated by a scale factor of 3. What is the area of the image? A. 13 B. 30 C. 90 D. 100

29. An exterior angle of a triangle equals the sum of its two remote interior angles, 40° and 75°. What is the exterior angle? A. 65° B. 105° C. 115° D. 125°

30. Two similar figures have corresponding sides in a ratio of 5 : 8. A side of the smaller is 15. What is the matching side of the larger? A. 18 B. 21 C. 24 D. 27

31. *(Griddable)* The base of a cylinder is a circle with radius 5. What is the base area B? ($\pi \approx 3.14$) ☐

32. Which transformation **changes** the size of a figure? A. Translation B. Reflection C. Rotation D. Dilation by factor 4

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 5, 7, 9, 11. ☐

34. A trend line is modeled by $y = 2x + 1$. Predict y when $x = 12$. A. 13 B. 23 C. 25 D. 27

35. On a scatterplot, points rise from lower-left to upper-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which pattern suggests the data are **linear**? A. A U-shaped curve B. A circular cluster C. A straight rising path D. Random scatter

37. *(Griddable)* Find the simple interest on $1,500 at 4% for 5 years. ☐

38. For the same amount and rate, which loan length costs the **most** total interest? A. 1 year B. 2 years C. 4 years D. 6 years

39. How do small, regular investments behave over many years? A. Shrink B. Stay equal C. Grow, especially with compound interest D. Disappear

40. College costs $20,000. A family contributes $8,000 and saves the rest over 24 months. How much per month? A. $400 B. $500 C. $600 D. $750

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Mathematics · Practice Test 4

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: ⅞, 0.88, 89%, √0.8. A. ⅞, 0.88, 89%, √0.8 B. √0.8, 89%, 0.88, ⅞ C. 0.88, ⅞, 89%, √0.8 D. 89%, ⅞, 0.88, √0.8

2. Between which two whole numbers does √150 lie? A. 11 and 12 B. 12 and 13 C. 13 and 14 D. 74 and 76

3. *(Griddable)* Write 0.000072 in scientific notation. ☐

4. Which number is **irrational**? A. √24 B. √25 C. ⅖ D. 0.1

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−3, 4) and (1, −4)? A. 2 B. −2 C. ½ D. −½

6. A proportional relationship passes through the origin and (6, 9). What is the unit rate? A. 1.5 B. 2 C. 3 D. 6

7. *(Equation editor)* A savings account starts with $40 and grows $15 each week. Write an equation for the balance y after x weeks. ☐

8. Which relation is a **function**? A. {(1, 2), (1, 5), (3, 4)} B. {(2, 3), (4, 3), (6, 5)} C. {(0, 1), (0, 4), (2, 6)} D. {(3, 1), (3, 9), (5, 2)}

9. Which equation represents a **non-proportional** relationship? A. y = 5x + 2 B. y = 6x C. y = ¼x D. y = 9x

10. *(Griddable)* For a direct variation, y = 54 when x = 6. What is y when x = 8? ☐

11. Which table shows a **proportional** relationship? A. (1, 5)(2, 10)(3, 15) B. (1, 3)(2, 7)(3, 11) C. (1, 2)(2, 5)(3, 9) D. (2, 9)(3, 13)(4, 17)

12. *(Griddable)* Solve for x: 8x + 3 = 5x + 27. ☐

13. "Nine more than twice a number equals five times the number minus six." Which equation models this? A. 2x + 9 = 5x − 6 B. 9 − 2x = 5x − 6 C. 2x − 9 = 5x + 6 D. 9x + 2 = 5x − 6

14. Two lines intersect at (−2, −5). What is the solution of the system? A. x = −5, y = −2 B. x = −2, y = −5 C. x = −7 D. No solution

15. The slope of a line is the same everywhere because the right triangles drawn to its points are — A. congruent B. **similar** C. obtuse D. equilateral

16. Which situation is **proportional**? A. $45 plus $20 per hour B. 3 apples for $1.50 at a steady rate C. 60 L losing 6 L/min D. $9 fee plus $4 each

17. Which situation can be modeled by $7x + 5 = 3x + 25$? A. Two savers start with $5 and $25, saving $7 and $3 per week; when equal? B. 7 times 5 C. A circle's area D. 5 plus 25

18. A phone plan costs $10 plus $0.05 per text. Which equation gives the cost y for x texts? A. $y = 10x + 0.05$ B. $y = 0.05x + 10$ C. $y = 10.05x$ D. $y = 0.05x - 10$

19. *(Select TWO)* Which statements about $y = -2x + 8$ are true? A. The slope is -2 B. The slope is 8 C. The y-intercept is 8 D. The y-intercept is -2

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 6 and height 5. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 282.6 C. 565.2 D. 1130.4

21. A sphere has radius 6. What is its volume? ($\pi \approx 3.14$) A. 150.72 B. 452.16 C. 904.32 D. 1808.64

22. A cylinder has radius 5 and height 4. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 219.8 C. 282.6 D. 314

23. *(Griddable)* A right triangle has legs of 20 and 21. Find the hypotenuse. ☐

24. What is the distance between $(-2, -1)$ and $(4, 7)$? A. 6 B. 8 C. 10 D. 14

25. The point $(-4, 6)$ is dilated by a scale factor of ½ about the origin. What is the image? A. $(-2, 3)$ B. $(-8, 12)$ C. $(-2, 6)$ D. $(-4, 3)$

26. The point $(3, -8)$ is reflected over the x-axis. What is the image? A. $(-3, -8)$ B. $(3, 8)$ C. $(-3, 8)$ D. $(8, 3)$

27. The point $(5, 2)$ is rotated 90° counterclockwise about the origin. What is the image? A. $(-2, 5)$ B. $(2, -5)$ C. $(-5, -2)$ D. $(2, 5)$

28. The angles of a triangle measure $2x$, $3x$, and $4x$. What is the **largest** angle? A. 20° B. 40° C. 60° D. 80°

29. A rectangle measures 6 by 4. It is dilated by a scale factor of ½. What is the area of the image? A. 3 B. 6 C. 12 D. 24

30. Two similar triangles have corresponding sides in a ratio of 4 : 7. A side of the smaller is 12. What is the matching side of the larger? A. 18 B. 20 C. 21 D. 24

31. In a right triangle, the square on the hypotenuse equals — A. the sum of the squares on the two legs B. the difference of the legs C. twice the longer leg D. the product of the legs

32. Which transformation **preserves** congruence? A. Dilation by factor 2 B. Dilation by factor 3 C. Rotation of 90° D. Dilation by factor ½

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 6, 10, 10, 14. ☐

34. A trend line is modeled by $y = 0.5x + 3$. Predict y when $x = 20$. A. 10 B. 13 C. 16 D. 23

35. A scatterplot shows points with no clear direction or pattern. What association is shown? A. Positive linear B. Negative linear C. No association D. Strong linear

36. Which pattern suggests a **non-linear** association? A. A straight rising line B. A straight falling line C. A curved path D. Points on one line

37. *(Griddable)* Find the simple interest on $2,000 at 3% for 6 years. ☐

38. Two accounts hold the same amount at the same rate for the same time. Which earns **more**? A. The simple-interest account B. The compound-interest account C. They earn the same D. Neither earns interest

39. What helps savings grow the most over time? A. Spending it quickly B. Saving once, then stopping C. Regular saving plus compound interest D. Keeping cash at home

40. College costs $24,000. A family contributes $9,000 and saves the rest over 30 months. How much per month? A. $450 B. $500 C. $600 D. $750

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Part 2 · Reading Language Arts

How Grade 8 RLA Is Built

The Grade 8 Reading Language Arts test has about **45 questions worth 56 points** in two sections, plus written responses. This book teaches **every reading and writing skill the test measures.**

Section	What It Covers	On the Test
Reading	Vocabulary, comprehension, literary elements, genres, author's craft; includes **2 Short Constructed Responses**	~28 Q / ~30 pts
Writing — Revising & Editing	Organizing and improving writing; grammar and conventions; includes **1 Extended Constructed Response**	~17 Q / ~26 pts

Three Kinds of Tasks

1. **Multiple-choice and tech-enhanced questions** about passages you read.
2. **Short Constructed Responses (SCR)** — a few written sentences with text evidence (2 points each).
3. **Extended Constructed Response (ECR)** — a full essay (10 points), tied to a reading passage.

Genres You Will Read

- **Literary:** fiction, literary nonfiction, poetry, drama
- **Informational / argumentative:** informational, argumentative, persuasive
- **In revising & editing:** correspondence (letters, emails)

> **Key Idea:** Reading STAAR is not about memorizing stories — it is about **using evidence.** Almost every question can be answered by pointing to a specific place in the text. Read closely and back up every answer with proof.

Genre Guide

STAAR passages come from many genres, and each has its own features to notice. Knowing the genre tells you what to look for. (Standards 8A–8E.)

Literary Genres

- **Fiction ○ (8A)** — made-up stories: realistic, historical, mystery, adventure, humor, fantasy, science fiction, short stories. *Notice:* characters, setting, conflict, plot, theme.
- **Literary Nonfiction** — true stories told with literary craft (memoir, personal essay). *Notice:* the real experience and its message.
- **Poetry ○ (8B)** — lines and stanzas; forms include epic, lyric, humorous. *Notice:* **graphical elements** such as line length and punctuation, plus rhythm, rhyme, imagery.
- **Drama ○ (8C)** — plays organized in **acts and scenes** with dialogue and stage directions. *Notice:* how acts and scenes move the action.

Informational and Argumentative Genres

- **Informational (8D)** — explains a topic. *Notice:* the **controlling idea/thesis** with evidence ★; features like **footnotes, endnotes, citations** ○; the **organizational pattern** ○.
- **Argumentative (8E)** — makes a claim and defends it. *Notice:* the **claim** ★, the **counter-argument** ★, and the **intended audience** ○.
- **Persuasive** — tries to convince. *Notice:* appeals, word choice, calls to action.

For Revising & Editing

- **Correspondence** — letters and emails, friendly or business. *Notice:* greeting, purpose, tone, closing.

> **Strategy:** The first thing to ask about any passage is *"What genre is this, and what is its purpose?"* A story wants you to follow characters and theme; an argument wants you to weigh a claim and its evidence.

Reading Strategy Lessons

These lessons teach the reading skills STAAR tests again and again. Each is tagged with its TEKS and marked ★ **Readiness** or ○ **Supporting**.

Part A · Vocabulary (2A ○, 2B ★, 2C ○)

- **2B · Context clues** ★ — Read the words around an unfamiliar word: definitions, examples, synonyms, antonyms, and the sentence's sense point to the meaning.
- **2A · Reference tools** ○ — A dictionary/glossary gives meaning, syllables, pronunciation, origin, part of speech. (A dictionary is available on the real test.)
- **2C · Greek & Latin roots** ○ — **ast** (star → astronomy), **qui** (rest → tranquil), **path** (feeling → sympathy), **mand/mend** (order → command), **duc** (lead → conduct).

Worked example. "The *arid* desert had not seen rain in years." *Arid* = very dry — the context shows it. **Try It.** "His *gregarious* nature made him the center of every party." *(Outgoing/sociable.)*

Part B · Comprehension (5C ○, 5E ★, 5F ★, 5G ★, 5H ★)

- **5F · Inference + evidence** ★ — A smart conclusion from text clues + prior knowledge; always find the supporting words.
- **5G · Key ideas** ★ — Separate the main point from small details. **5H · Synthesize** ★ — Combine ideas (one text or across two) into new understanding.
- **5E · Make connections** ★ — Link text to other texts, the world, and experience. **5C · Predictions** ○ — Predict from genre and features, then confirm or correct.

Worked example. A character "slammed the door and refused to speak" → **infer** she is angry; the evidence is her actions. **Try It.** Sky darkens, wind rises, people hurry indoors → *a storm is coming.*

Part C · Response to Reading (6C ★, 6D ★)

- **6C · Text evidence** ★ — Support every answer with a specific word, phrase, or sentence. This is the heart of STAAR reading.
- **6D · Paraphrase & summarize** ★ — Restate in your own words (paraphrase) or give a short, ordered version of the main points (summarize), keeping meaning accurate.

Strategy: For "Which sentence **best** supports…?" decide your answer first, then find the line that *directly* proves it — not one that is merely true.

Part D · Literary Elements (7A ○, 7B ★, 7C ★, 7D ○)

- **7B · Character motivation** ★ — Why characters act, and how that drives events and the conflict's resolution.
- **7C · Non-linear plot** ★ — **Flashback**, **foreshadowing**, subplots, parallel plots — vs. straight linear order.

- **7A · Theme** ○ — The message, developed through characters and events. **7D · Setting** ○ — When/where, and how it shapes values and beliefs.

Try It. A story opens at a funeral, then jumps back to the character's childhood. What device? *(Flashback.)*

Part E · Author's Purpose and Craft (9A ★, 9B–9G ○)

- **9A · Purpose & message** ★ — To inform, persuade, or entertain — and the message within.
- **9B · Text structure** ○ — Cause–effect, compare–contrast, sequence, problem–solution. **9C · Print/graphic features** ○ — Headings, captions, charts, and why.
- **9D · Figurative language** ○ — Simile, metaphor (incl. extended metaphor), personification. **9E · Devices** ○ — Irony and multiple points of view.
- **9F · Mood, voice, tone** ○ — The feeling, the author's personality, the attitude toward the subject. **9G · Rhetoric & fallacies** ○ — Analogy, juxtaposition; bandwagon, circular reasoning.

Worked example (tone). "cramped, gloomy, endless" → a **negative, heavy tone.** **Try It.** "Everyone is switching to BrightSmile — don't be left behind!" Which fallacy? *(Bandwagon appeal.)*

Reading Skills Checklist

- ☐ 2A–C Vocabulary · ☐ 5C–H Comprehension · ☐ 6C–D Response
- ☐ 7A–D Literary elements · ☐ 9A–G Author's craft

Revising and Editing Refresher

In the Writing section you fix and improve short passages. **Revising** makes writing clearer and better organized; **editing** corrects grammar and conventions. (Standards 10B, 10C, 10D.)

Revising — Making Writing Better (10B ★, 10C ★)

- **10B · Structure & development ★** — A strong piece has an **introduction, transitions, coherent paragraphs, and a conclusion** (10B·i) and develops ideas with **specific facts, details, examples** (10B·ii).
- **10C · Revising for quality ★** — Improve **clarity, development, organization, style, word choice, and sentence variety.** Combine choppy sentences; cut repetition; choose precise words.

Worked example. "The dog was big. The dog was loud. The dog ran." → "The big, loud dog ran across the yard." (sentence variety)

Editing — Correcting Conventions (10D)

- **10D·i · Complete sentences ★** — Complete sentences with **subject-verb agreement**; avoid run-ons, comma splices, fragments.
- **10D·ii · Verb tense & voice ★** — Keep tenses consistent; use active/passive voice correctly.
- **10D·iii · Prepositions** ○ — Use correctly; don't let them confuse subject-verb agreement. **10D·iv · Pronoun-antecedent** ○ — Match pronoun to noun in number and gender.
- **10D·v · Capitalization** ○ **10D·vi · Punctuation** ○ — Commas in nonrestrictive phrases/clauses, **semicolons, colons, parentheses.**
- **10D·vii · Spelling ★** — Including confusables: **its/it's, affect/effect, there/their/they're, to/two/too.**

Worked example (agreement). "The box of markers *are* on the desk." → subject is *box* (singular): "The box of markers **is** on the desk." (the phrase *of markers* doesn't change the subject)

> **Watch Out:** *Its* = belonging to it; *it's* = it is. *Their* = belonging to them; *there* = a place; *they're* = they are. Tested often.

Try It. Fix: "Me and him walked to there house." → *(He and I walked to their house.)*

Short Constructed Response (SCR) Mastery

The Reading section includes **2 Short Constructed Responses**, each worth **2 points.** You write a few sentences that answer a question and back it up with **evidence from the text.** (Builds on response standards 6C and 6D.)

Score	What it means
2	A **correct, clear answer** fully supported with **accurate text evidence** and a brief explanation
1	A partially correct answer, or weak/vague/incomplete evidence
0	An incorrect answer, no text evidence, or blank/off-topic

Key Idea: Two points = **answer + evidence + explanation.** Leaving out the evidence is the most common way students lose a point.

The A-E-E Structure

1. Answer the question directly. **2. Evidence** — quote or paraphrase a specific detail. **3. Explain** how it supports your answer. Stems: *The text shows that… · For example, the author writes "…" · This proves that…*

Worked Example

Prompt: *How does the narrator feel about moving to a new town? Use evidence from the text.*

Full credit (2): The narrator feels **nervous and sad** about moving. The text says she "stared at the empty walls of her old room and felt her stomach tighten." This shows she is anxious because her familiar home is gone. *— clear answer + quotation + explanation.*

Partial (1): The narrator feels bad about moving because she doesn't want to go. *— reasonable, but no text evidence or explanation.*

No credit (0): I would feel excited to move to a new place. *— personal opinion, not the narrator's feelings; no evidence.*

Strategy: Keep it to three or four focused sentences. Always include one specific detail or short quotation. Answer the exact question asked — don't drift into your own opinion unless the prompt asks for it.

Extended Constructed Response (ECR) Mastery

The Writing section includes **1 Extended Constructed Response** — a full multi-paragraph essay worth **10 points**, tied to a reading passage. It is the single biggest-point task on the RLA test.
(Composition standards 11B informational, 11C argumentative, 11D correspondence; and 6B comparing sources.)

What the Essay Must Have

- A clear **central idea or claim** (thesis).
- **Coherent organization** — introduction, body paragraphs, conclusion.
- **Sufficient development** — specific **evidence** and explanation, not general statements.
- **Effective language and conventions** — varied sentences, precise words, correct grammar.

> **Key Idea:** Points come from a clear thesis, organized paragraphs, and **specific evidence** explained in your own words. A blank or one-paragraph answer cannot score high.

A 4-Step Essay Plan

1. **Plan (2–3 min).** Decide your thesis; jot 2–3 supporting points and the evidence for each.
2. **Introduction.** Hook + topic + **thesis/claim.**
3. **Body paragraphs (2–3).** One point each: state it → give **evidence** → **explain** it. Use transitions.
4. **Conclusion.** Restate the thesis in fresh words; leave a final thought.

Argumentative tip: Name the **counter-argument** and answer it ("Some may say…, but…").

Annotated Sample (Argumentative, shortened)

Prompt: *Should students have longer recess?*

Many schools are cutting recess, but students actually learn better with more of it. *(claim)* First, recess gives the brain a break. The article states that "students who move during the day focus longer in class." *(evidence)* This shows play is not wasted time. *(explanation)* In addition, recess builds friendships and reduces stress. Some argue recess takes away learning minutes, **but** rested students finish work faster. *(counter-argument answered)* For these reasons, schools should give students longer recess. *(conclusion)*

Score band	What it looks like
High (8–10)	Clear thesis, well organized, specific evidence explained, strong conventions

Middle (4–7)	A thesis and some organization, but thin evidence or several errors
Low (1–3)	Unclear or missing thesis, little development, many errors
0	Blank, off-topic, or too brief to score

Strategy: Plan for a couple of minutes before writing — an organized essay almost always scores higher than a longer, messy one. Always use **evidence from the passage**, and leave a minute to check spelling and punctuation.

Reading Language Arts · Practice Test 1

Read each passage, then answer the questions that follow. Use evidence from the texts.

Passage 1 — The Audition

(1) Maya's fingers hovered over the piano keys. The recital hall was empty except for Ms. Alvarez, who waited with a clipboard and a patient smile. For weeks Maya had practiced the same piece until her hands ached, yet now every note seemed to hide from her.

(2) "Whenever you're ready," Ms. Alvarez said.

(3) Maya's heart thudded. She thought of the school talent show two years earlier, when she had frozen on stage and walked off to a silent crowd. The memory still stung. She almost reached for her bag, ready to leave.

(4) Then she remembered her grandmother's words: "Courage isn't the absence of fear, Maya. It's playing anyway." Maya took a slow breath, closed her eyes, and began.

(5) The first notes trembled, but by the second measure her hands remembered what her mind had forgotten. The music filled the empty hall, rising and falling like waves. When the final chord faded, Maya opened her eyes. She was smiling.

(6) "That," said Ms. Alvarez, setting down her clipboard, "was worth the wait."

(7) Maya walked out lighter than she had entered. The fear had not vanished—but she had played anyway, and that had made all the difference.

1. In paragraph 1, the word *hovered* most nearly means — A. rested lightly B. pressed hard C. slipped off D. disappeared

2. Why does Maya almost reach for her bag? A. She forgot her music B. She recalls freezing at the talent show C. She dislikes Ms. Alvarez D. She is bored

3. The grandmother's words suggest that courage means — A. never feeling afraid B. avoiding challenges C. acting in spite of fear D. practicing more

4. Which statement **best** expresses a theme of the passage? A. Talent is something you are born with B. Facing your fear leads to growth C. Music is too hard for most people D. Teachers are strict

5. The mood at the end of the passage is — A. tense and fearful B. calm and hopeful C. angry D. confused

6. Which sentence **best** shows that Maya has overcome her fear? A. "Her hands ached." B. "Every note seemed to hide from her." C. "She had played anyway, and that had made all the difference." D. "The recital hall was empty."

7. Paragraph 3, about the talent show two years earlier, is an example of — A. foreshadowing B. flashback C. climax D. setting

SCR 1. *(2 points)* How does Maya change from the beginning to the end of the passage? Support your answer with **evidence from the text.**

Passage 2 — Cities Buzz Again

(1) Not long ago, honeybees were a rare sight in big cities. Today, hives sit on rooftops, in community gardens, and even atop hotels. Urban beekeeping is growing fast—and cities are better for it.

(2) Bees are pollinators. As they gather nectar, they move pollen from flower to flower, helping plants produce fruits, vegetables, and seeds. A single hive can pollinate millions of flowers in a day. In a city, that means healthier parks, gardens, and street trees.

(3) Urban hives can even be healthier than rural ones. City gardens bloom with many kinds of flowers, so bees enjoy a varied diet. And because many city gardeners avoid heavy pesticides, urban bees may face fewer harmful chemicals than bees near large farms.

(4) Beekeeping also connects people. Neighbors who tend a shared hive learn about nature and work together. Schools use rooftop hives to teach science. Local honey, sold at markets, supports small businesses.

(5) Of course, beekeeping takes training. Beekeepers must know how to handle bees safely and keep hives healthy. But with care, a city hive brings sweet rewards—for people, plants, and the bees themselves.

8. The central idea of the passage is that — A. bees are dangerous in cities B. urban beekeeping benefits cities C. honey is too expensive D. farms no longer have bees

9. The author's main purpose is to — A. entertain with a story B. inform readers about the benefits of urban beekeeping C. sell honey D. warn people to stay away from bees

10. Paragraph 3 is organized mainly by — A. comparing urban and rural hives B. telling a story in order C. listing dangers D. defining a single word

11. As used in paragraph 2, *pollinators* are organisms that — A. make honey only B. move pollen between flowers C. build hives D. sting people

12. Which detail **best** supports the idea that city bees may be healthier than rural bees? A. "hives sit on rooftops" B. "City gardens bloom with many kinds of flowers, so bees enjoy a varied diet." C. "Beekeeping takes training." D. "Local honey supports small businesses."

13. From paragraph 3, the reader can infer that heavy pesticide use — A. helps bees B. has no effect on bees C. can be harmful to bees D. makes more honey

14. Which is the **best** summary of the passage? A. Bees sting, so cities ban them. B. Urban beekeeping is growing and benefits plants, people, and bees, though it takes training. C. Honey tastes sweet. D. Schools have rooftops.

SCR 2. *(2 points)* Explain **one** way urban beekeeping helps a city. Use **evidence from the passage.**

Paired Set — Rooftop Garden & Green Roofs

Text A — "Rooftop Garden" (poem)

Above the traffic's restless hum,
a quiet green begins to climb—
tomatoes red as stoplights,
basil bright with summer time.
Here the city tilts its face
toward sun and rain and seed,
and proves that even concrete
can hold what gardens need.

Text B — "Green Roofs"

Rooftop gardens, or "green roofs," do more than grow food. They soak up rainwater, cool buildings in summer, and give insects and birds a place to rest. Some cities now pay building owners who plant them.

15. In Text A, "tomatoes red as stoplights" is an example of — A. metaphor B. simile C. personification D. alliteration

16. In Text A, "the city tilts its face toward sun" is an example of — A. simile B. personification C. hyperbole D. irony

17. The main idea of Text B is that green roofs — A. are too expensive B. provide several benefits C. only grow food D. damage buildings

18. Both texts suggest that rooftop gardens — A. are impossible in cities B. bring nature into the city C. should replace all parks D. are only for growing food

19. Which idea appears in Text B but **not** in Text A? A. Plants grow on rooftops B. Green roofs soak up rainwater and cool buildings C. Gardens need sun D. Cities are busy

20. Text A is written mainly to — A. inform with facts B. express a feeling about city gardens C. argue a point D. give step-by-step instructions

Revising — "A Recycling Program"

(1) Our school should start a recycling program. (2) Right now, we throw away hundreds of bottles every week. (3) A recycling program would help the environment. (4) It would also teach students good habits. (5) Some students might forget to use the bins at first. (6) But with reminders, it would become normal.

21. Which sentence could **best** be added after sentence 2 to develop the idea with a detail? A. "Recycling is good." B. "Last month our class counted over 400 plastic bottles in the trash." C. "I like our school." D. "The bins are blue."

22. Which transition would **best** begin sentence 4? A. However, B. For example, C. In addition, D. Although

23. What is the **best** way to combine sentences 3 and 4? A. ...environment, it also teaches good habits. B. A recycling program would help the environment and teach students good habits. C. ...environment, and, teaching good habits. D. Helping the environment and good habits a recycling program.

24. Which sentence **best** states the writer's main claim (thesis)? A. Sentence 1 B. Sentence 2 C. Sentence 5 D. Sentence 6

Editing — "The Science Museum"

(1) My family and me visited the science museum last Saturday. (2) The exhibit about space were amazing. (3) We seen a real moon rock and a model rocket. (4) Its hard to believe how big the universe is.

25. What change should be made in sentence 1? A. Change "me" to "I" B. Change "visited" to "visit" C. Change "last" to "latest" D. No change

26. What change should be made in sentence 2? A. Change "were" to "was" B. Change "about" to "of" C. Change "exhibit" to "exhibits" D. No change

27. What change should be made in sentence 3? A. Change "seen" to "saw" B. Change "real" to "really" C. Change "rock" to "rocks" D. No change

28. What change should be made in sentence 4? A. Change "Its" to "It's" B. Change "believe" to "belief" C. Change "how" to "how is" D. No change

Extended Constructed Response (ECR) — *(10 points)*

Passage 2, "Cities Buzz Again," describes the benefits of urban beekeeping. **Should your community start an urban beekeeping or rooftop-garden program?** Write a multi-paragraph **argumentative essay** that states your position and supports it with reasons and evidence, including evidence from the passage. Organize your essay with an introduction, body paragraphs, and a conclusion, and use correct grammar and spelling.

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Reading Language Arts · Practice Test 2

Read each passage, then answer the questions that follow. Use evidence from the texts.

Passage 1 — Opening Night (drama)

Backstage at a school theater. PRIYA paces. JAMAL leans against a prop table.

PRIYA: *(pacing)* I can't do this, Jamal. There are a hundred people out there.

JAMAL: A hundred and two. I counted. *(grinning)*

PRIYA: You're not helping.

JAMAL: *(serious)* You've known these lines for a month. You said them in your sleep—your sister texted me.

PRIYA: *(half-laughing)* She did not.

JAMAL: The point is, you're ready. The nerves just mean you care.

(The STAGE MANAGER waves from the wings.)

STAGE MANAGER: Priya—thirty seconds.

PRIYA: *(takes a deep breath)* Okay. Okay. *(She steps toward the light, then turns back.)* Jamal? Thanks.

JAMAL: Break a leg.

(Priya walks onstage. The backstage lights dim as applause rises.)

1. The words in parentheses, such as *(pacing)* and *(grinning)*, are — A. dialogue B. stage directions C. titles D. settings

2. Why does Jamal joke that he counted "a hundred and two" people? A. to frighten Priya B. to lighten her nerves C. to leave the theater D. to brag

3. Priya's pacing and words show that she feels — A. bored B. confident C. nervous D. angry

4. A theme of the scene is that — A. friends help us face our fears B. acting is easy C. counting calms everyone D. plays run too long

5. As used in the scene, "Break a leg" means — A. get hurt B. good luck C. stop now D. run away

6. Which line **best** shows Priya gathering her courage? A. "I can't do this" B. "You're not helping" C. "*(takes a deep breath)* Okay. Okay." D. "There are a hundred people out there"

7. The stage manager's entrance mainly serves to — A. end the play B. move the action toward Priya's entrance C. begin a new conflict D. describe the setting

SCR 1. *(2 points)* How does Jamal help Priya in this scene? Support your answer with **evidence from the text.**

Passage 2 — Phones in School (argument)

(1) Should students use phones during the school day? Many schools say no—and they are right. Phones in class do more harm than good.

(2) First, phones distract. Studies show that students who keep phones nearby check them dozens of times an hour, breaking focus during lessons. Even a buzzing pocket pulls attention away from learning.

(3) Second, phones can harm how students treat one another. Cyberbullying often happens during school hours, and constant texting can crowd out real conversation at lunch.

(4) Some argue that phones are useful for emergencies and research. That is a fair point. But schools already have office phones for emergencies and classroom computers for research. The benefits do not outweigh the costs.

(5) A phone-free school day helps students focus, connect, and learn. For most schools, putting phones away is a smart choice.

8. The author's main **claim** is that — A. phones help learning B. phones should be put away during school C. schools need more computers D. emergencies never happen

9. In paragraph 4, the author brings up the idea that phones help in emergencies in order to — A. agree with it completely B. address and answer the opposing view C. change the subject D. confuse the reader

10. The author's purpose is to — A. entertain with a story B. persuade readers that phones should be put away C. describe a phone D. explain how phones are built

11. Which detail **best** supports the idea that phones distract? A. "schools already have office phones" B. "students check them dozens of times an hour" C. "a phone-free school day" D. "conversation at lunch"

12. This passage is most likely written for — A. toddlers B. school leaders, students, and parents C. scientists only D. airline pilots

13. The author builds the argument mainly by — A. telling a personal story B. giving reasons and answering a counter-argument C. listing dictionary definitions D. using rhyme

14. Which is the **best** summary? A. Phones are fun. B. The author argues phones should be put away during school because they distract and harm relationships, and their benefits can be met in other ways. C. Schools have computers. D. Texting is common.

SCR 2. *(2 points)* According to the author, why should phones be put away during the school day? Use **evidence from the passage.**

Paired Set — Teen Sleep

Text A — "Why Teens Need Sleep"

Teenagers need about 8 to 10 hours of sleep each night, yet most get far less. During puberty, the body's internal clock shifts later, so teens naturally feel awake at night and sleepy in the morning. Sleep helps memory, mood, and health. Without enough, students struggle to focus and remember what they learn.

Text B — "Later Start Times"

Some schools have tried starting classes later in the morning. The results are promising: students report better attendance, higher grades, and fewer car accidents among teen drivers. Sleep experts say a start time of 8:30 a.m. or later fits teens' natural sleep patterns.

15. The main idea of Text A is that — A. teens need enough sleep for health and learning B. teens sleep too much C. clocks are broken D. mornings are the best time to study

16. Text B suggests that later start times — A. hurt grades B. have several benefits for teens C. are impossible to try D. cause more accidents

17. Both texts support the idea that — A. teens need less sleep B. teen sleep affects school success C. schools should start earlier D. sleep does not matter

18. Which idea appears in Text B but **not** in Text A? A. Teens need sleep B. Later start times improve attendance and grades C. Sleep helps memory D. The body clock shifts

19. Text A is organized mainly by — A. explaining a cause (the body clock) and its effects B. telling a story in order C. listing steps D. comparing two schools

20. Both texts are written mainly to — A. entertain B. inform readers about teen sleep C. sell a product D. argue against school

Revising — "A School Garden"

(1) Our class wants to build a school garden. (2) A garden would give us fresh vegetables. (3) It would teach us about science. (4) Working outside is fun. (5) We would need soil, seeds, and tools. (6) With help from teachers, we can make it happen.

21. Which sentence could **best** be added after sentence 3 to develop it with a detail? A. "Science is a subject." B. "For example, we could study how plants grow and measure them each week." C. "I like gardens." D. "Tools are useful."

22. Which transition would **best** begin sentence 4? A. However, B. In addition, C. Because D. Finally,

23. What is the **best** way to combine sentences 2 and 3? A. A garden would give us fresh vegetables, it teaches science. B. A garden would give us fresh vegetables and teach us about science. C. Fresh vegetables and science, a garden. D. A garden, giving vegetables, science.

24. Which sentence **best** states the writer's main idea (thesis)? A. Sentence 1 B. Sentence 4 C. Sentence 5 D. Sentence 6

Editing — "The Field Trip"

> (1) Our class go to the aquarium every year. (2) Last year, me and my friends saw a giant octopus. (3) It's arms were longer than I am tall. (4) We learned alot about ocean animals.

25. What change should be made in sentence 1? A. Change "go" to "goes" B. Change "class" to "classes" C. Change "every" to "ever" D. No change

26. What change should be made in sentence 2? A. Change "me and my friends" to "my friends and I" B. Change "saw" to "seen" C. Change "giant" to "gigantic" D. No change

27. What change should be made in sentence 3? A. Change "It's" to "Its" B. Change "longer" to "long" C. Change "am" to "is" D. No change

28. What change should be made in sentence 4? A. Change "alot" to "a lot" B. Change "learned" to "learn" C. Change "ocean" to "Ocean" D. No change

Extended Constructed Response (ECR) — *(10 points)*

Using **Text A ("Why Teens Need Sleep")** and **Text B ("Later Start Times")**, write a multi-paragraph **informational essay** that explains **why teenagers need enough sleep and how schools could help.** Use evidence from **both** texts. Organize your essay with an introduction, body paragraphs, and a conclusion, and use correct grammar and spelling.

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Reading Language Arts · Practice Test 3

Read each passage, then answer the questions that follow. Use evidence from the texts.

Passage 1 — The First Catch (literary nonfiction)

(1) I was nine the summer my grandfather taught me to fish. Every morning we rowed to the middle of the gray lake and waited. And waited. For three days I caught nothing but boredom.

(2) "Fishing isn't about catching," Grandpa said on the fourth morning, reeling in another empty line. "It's about paying attention."

(3) I didn't understand him. I wanted a fish—something to hold up and photograph. But I kept casting, because he kept casting.

(4) On the fifth day, my line went tight. My heart leaped. I pulled too hard, and the fish slipped free. I groaned, but Grandpa only smiled. "Now you're paying attention," he said.

(5) The next cast, I waited. I felt the smallest tug, counted to three, and lifted gently. A silver fish broke the surface, flashing in the sun. I had done it.

(6) Years later, I still remember what the lake taught me that summer. Patience, it turns out, is its own kind of catch.

1. In paragraph 1, "nothing but boredom" suggests the narrator felt — A. excited B. impatient and bored C. afraid D. proud

2. The narrator keeps casting mainly because — A. he already loves fishing B. his grandfather keeps casting C. he is forced to D. he wants to row the boat

3. When Grandpa says "Now you're paying attention," he means the narrator — A. caught a fish B. is finally focused and learning C. should go home D. is tired

4. The central lesson of the memoir is that — A. fishing is boring B. patience is valuable C. grandfathers are wise D. lakes are gray

5. The author wrote this mainly to — A. inform about fish species B. share a meaningful personal experience and its lesson C. sell fishing gear D. argue a point

6. Which line **best** shows that the narrator learned patience? A. "I wanted a fish" B. "I pulled too hard, and the fish slipped free" C. "The next cast, I waited... and lifted gently" D. "we rowed to the middle"

7. "Patience... is its own kind of catch" is an example of — A. simile B. metaphor C. alliteration D. onomatopoeia

SCR 1. *(2 points)* What does the narrator learn by the end of the memoir? Support your answer with **evidence from the text.**

Passage 2 — Save Our Library (persuasive)

(1) Our school library is more than a room full of books. It is the heart of our school—and it needs our help.

(2) Every day, students visit the library to read, study, and find a quiet place to think. For some, it is the only place they can borrow books they could never afford to buy. Closing it, or cutting its hours, would leave these students behind.

(3) Imagine a school where no one reads for joy, where research means guessing instead of learning. Is that the school we want?

(4) We can act now. Sign the petition, attend the board meeting, and tell the district that our library matters. Together, our voices are loud enough to save it.

(5) Don't wait until the shelves are empty. Stand up for your library today.

8. The author's purpose is to — A. inform about library hours B. persuade readers to support the library C. tell a story D. describe a book

9. The author's main message is that — A. libraries are outdated B. the community should act to save the library C. books are expensive D. students should study more

10. "Is that the school we want?" is a — A. statistic B. rhetorical question meant to persuade C. definition D. stage direction

11. Which detail **best** supports the idea that the library helps students who cannot buy books? A. "a room full of books" B. "the only place they can borrow books they could never afford to buy" C. "attend the board meeting" D. "the heart of our school"

12. "Stand up for your library today" is an example of — A. a call to action B. a counter-argument C. a citation D. a simile

13. The author most likely wrote this because the library is — A. brand new B. at risk of closing or cuts C. too big D. never used

14. Which is the **best** summary? A. The library has books. B. The author urges readers to take action to save a valued school library that some students depend on. C. Board meetings are long. D. Reading is fun.

SCR 2. *(2 points)* Why does the author believe the library is important? Use **evidence from the passage.**

Paired Set — The Old Oak & Why Trees Matter

Text A — "The Old Oak" (poem)

For a hundred years it stood,
arms open to the sky,
shade for the weary traveler,
home where the songbirds fly.
It asked for nothing, gave for free—
the patient, giving tree.

Text B — "Why Trees Matter"

Trees do quiet, important work. They clean the air by taking in carbon dioxide and releasing oxygen. Their roots hold soil in place and soak up rainwater, preventing floods. Their shade cools neighborhoods and lowers energy costs. Protecting old trees protects us, too.

15. In Text A, calling the tree "patient" and "giving" is an example of — A. personification B. simile C. irony D. hyperbole

16. Text A is written in — A. paragraphs B. lines and a stanza with rhyme C. a script D. a chart

17. The main idea of Text B is that trees — A. are pretty B. provide important environmental benefits C. live a long time D. give birds homes

18. Both texts present trees as — A. useless B. valuable and worth protecting C. dangerous D. temporary

19. Which idea appears in Text B but **not** in Text A? A. Trees give shade B. Trees clean the air and prevent floods C. Trees are homes for birds D. Trees are old

20. Text A mainly aims to — A. inform with data B. express appreciation for a tree C. argue for a law D. give directions

Revising — "A Class Trip"

(1) Our class should visit the history museum. (2) The museum has exhibits about our state. (3) We would learn a lot. (4) It is close to our school. (5) Some students have never been. (6) A trip would be fun and educational.

21. Which sentence could **best** be added after sentence 2 to develop it with a detail? A. "Museums are nice." B. "For example, one exhibit shows tools and clothing from a hundred years ago." C. "I like trips." D. "It is a building."

22. Which transition would **best** begin sentence 4? A. However, B. In addition, C. Because D. But

23. What is the **best** way to combine sentences 3 and 4? A. We would learn a lot, it is close. B. We would learn a lot, and the museum is close to our school. C. Learn a lot and close museum. D. A lot of learning, close, museum.

24. Which sentence **best** states the writer's main idea (thesis)? A. Sentence 1 B. Sentence 3 C. Sentence 5 D. Sentence 6

Editing — "My Dog"

(1) My dog Rex run to the door every morning. (2) He wait for me to take him outside. (3) Their is nothing he loves more than a walk. (4) After are walk, he takes a long nap.

25. What change should be made in sentence 1? A. Change "run" to "runs" B. Change "dog" to "dogs" C. Change "every" to "ever" D. No change

26. What change should be made in sentence 2? A. Change "wait" to "waits" B. Change "me" to "I" C. Change "him" to "he" D. No change

27. What change should be made in sentence 3? A. Change "Their" to "There" B. Change "nothing" to "no thing" C. Change "loves" to "love" D. No change

28. What change should be made in sentence 4? A. Change "are" to "our" B. Change "walk" to "walks" C. Change "nap" to "naps" D. No change

Extended Constructed Response (ECR) — *(10 points)*

Passage 2, "Save Our Library," argues that the school library should be saved. **Do you agree that schools should invest in their libraries?** Write a multi-paragraph **argumentative essay** that states your position and supports it with reasons and evidence, including evidence from the passage. Organize your essay with an introduction, body paragraphs, and a conclusion, and use correct grammar and spelling.

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Reading Language Arts · Practice Test 4

Read each passage, then answer the questions that follow. Use evidence from the texts.

Passage 1 — The Signal (science fiction)

(1) Nadia had built the radio herself, from parts she had scavenged over two years. Tonight, for the first time, it hummed to life.

(2) At first there was only static—the ordinary hiss of an empty sky. She almost switched it off. Then, beneath the noise, she heard it: a pattern. Three short tones, three long, three short. Again. And again.

(3) Her breath caught. That was no accident. Someone—or something—was sending a message.

(4) She thought of her father, who had spent his life insisting that we were not alone, and who had died before anyone believed him. She had promised to keep listening. She had nearly given up.

(5) Nadia grabbed her notebook and began to write down the pattern, her hand shaking. Whatever waited at the other end of that signal, she would answer it.

(6) Outside, the night sky stretched on, vast and silent—but to Nadia, it would never feel empty again.

1. In paragraph 1, *scavenged* most nearly means — A. bought new B. collected from discarded things C. broke apart D. painted

2. Nadia keeps listening mainly because — A. she is bored B. she promised her father C. the radio is loud D. she enjoys static

3. The repeating pattern of tones suggests — A. a broken radio B. a deliberate message C. a storm D. music

4. A theme of the story is that — A. technology is dangerous B. persistence can lead to discovery C. radios are outdated D. space is empty

5. The mood at the end of the passage is — A. hopeful and full of wonder B. angry C. bored D. fearful

6. Which line **best** shows that Nadia had almost quit? A. "she had built the radio herself" B. "She had nearly given up." C. "three short, three long, three short" D. "the night sky stretched on"

7. This passage is best described as — A. informational text B. science fiction C. a persuasive essay D. a poem

SCR 1. *(2 points)* How does the signal change the way Nadia feels about the night sky? Support your answer with **evidence from the text.**

Passage 2 — The Hidden World of Fungi (informational)

(1) When you picture a forest, you probably think of trees. But beneath your feet lies a hidden world that keeps the forest alive: fungi.

(2) Fungi are not plants or animals; they form their own kingdom. Most of a fungus is not the mushroom you see, but a web of tiny threads called mycelium, spreading through the soil.

(3) These threads connect to the roots of trees, trading nutrients for sugar. Through this network, trees can even share food and send warning signals to one another. Scientists sometimes call it the "wood wide web."

(4) Fungi also recycle. They break down fallen leaves and dead wood, returning nutrients to the soil so new plants can grow. Without fungi, forests would drown in their own waste.

(5) The next time you walk through the woods, remember: the most important part of the forest may be the part you cannot see.

8. The central idea of the passage is that — A. mushrooms are tasty B. fungi play a hidden but vital role in forests C. trees are tall D. soil is dirty

9. As used in paragraph 2, *mycelium* is — A. a mushroom cap B. a web of tiny threads in the soil C. a kind of tree D. an animal

10. Paragraph 4 is organized mainly by — A. explaining a function (recycling) and its result B. telling a story in order C. comparing two forests D. listing names

11. The author's purpose is to — A. persuade readers to eat mushrooms B. inform readers about the role of fungi C. tell a scary story D. sell soil

12. Which detail **best** supports the idea that fungi help trees? A. "fungi are not plants or animals" B. "These threads connect to the roots of trees, trading nutrients for sugar" C. "forests have trees" D. "walk through the woods"

13. From paragraph 4, the reader can infer that without fungi — A. forests would grow faster B. dead material would pile up C. trees would move D. soil would vanish

14. Which is the **best** summary? A. Forests have trees. B. Fungi form a hidden network that connects trees and recycles nutrients, making them vital to forests. C. Mushrooms grow in soil. D. Scientists like webs.

SCR 2. *(2 points)* Explain **one** way fungi help a forest. Use **evidence from the passage.**

Paired Set — Bike Lanes

Text A — "Build More Bike Lanes" (argument)

Cities should build more protected bike lanes. They make streets safer for everyone, cut traffic, and reduce pollution. When people feel safe riding, more of them leave their cars at home.

Text B — "By the Numbers"

A recent study found that streets with protected bike lanes had 40% fewer injuries for all road users. Cities that added bike lanes also saw a rise in local business as more people stopped to shop.

15. The claim in Text A is that — A. bikes are expensive B. cities should build more protected bike lanes C. cars are fast D. streets are wide

16. Text B mainly provides — A. a personal story B. data that supports bike lanes C. an opinion against bikes D. a poem

17. Together, the texts suggest that bike lanes — A. are useless B. benefit safety and communities C. only help cyclists D. cause traffic

18. Which specific evidence appears in Text B but **not** in Text A? A. bike lanes are safer B. the "40% fewer injuries" statistic C. cities have streets D. people ride bikes

19. Text A's main purpose is to — A. inform with statistics B. argue for building bike lanes C. entertain D. give directions

20. Text B strengthens Text A's argument mainly by — A. adding a story B. providing facts and numbers as evidence C. disagreeing with it D. using rhyme

Revising — "A Chess Club"

(1) Our school should start a chess club. (2) Chess teaches planning. (3) It helps students think ahead. (4) Anyone can learn. (5) We would meet once a week. (6) A chess club would be a great addition to our school.

21. Which sentence could **best** be added after sentence 3 to develop it with a detail? A. "Chess is old." B. "For example, players must think several moves ahead to win." C. "I like games." D. "Clubs meet."

22. Which transition would **best** begin sentence 4? A. However, B. In addition, C. Because D. Finally,

23. What is the **best** way to combine sentences 2 and 3? A. Chess teaches planning, it helps think ahead. B. Chess teaches planning and helps students think ahead. C. Planning and thinking, chess. D. Chess, planning, thinking ahead.

24. Which sentence **best** states the writer's main idea (thesis)? A. Sentence 1 B. Sentence 2 C. Sentence 4 D. Sentence 5

Editing — "The Concert"

(1) Last night, my sister and me went to a concert. (2) The band were amazing. (3) We clapped until our hands hurt. (4) It was the best night of my hole life.

25. What change should be made in sentence 1? A. Change "me" to "I" B. Change "went" to "gone" C. Change "night" to "nite" D. No change

26. What change should be made in sentence 2? A. Change "were" to "was" B. Change "band" to "bands" C. Change "amazing" to "amazed" D. No change

27. What change should be made in sentence 3? A. Change "clapped" to "clap" B. Change "our" to "are" C. Change "hands" to "hand" D. No change

28. What change should be made in sentence 4? A. Change "hole" to "whole" B. Change "best" to "bestest" C. Change "night" to "Night" D. No change

Extended Constructed Response (ECR) — *(10 points)*

Using **"The Hidden World of Fungi,"** write a multi-paragraph **informational essay** that explains **why fungi are important to forests.** Use evidence from the passage. Organize your essay with an introduction, body paragraphs, and a conclusion, and use correct grammar and spelling.

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Part 3 · Answer Keys & Explanations

Use the quick keys to score fast, then read the full explanations to learn from every question. "CR" marks a constructed-response item (SCR or ECR) scored with a rubric — see the SCR/ECR scored samples.

Quick Answer Keys — Mathematics

#	Test 1	Test 2	Test 3	Test 4	#	Test 1	Test 2	Test 3	Test 4
1	A	A	A	A	21	B	C	B	C
2	B	B	B	B	22	C	B	C	C
3	6.8×10^{4}	4.5×10^{-4}	5.2×10^{6}	7.2×10^{-5}	23	25	41	17	29
4	C	B	A	A	24	A	B	A	C
5	B	B	A	B	25	A	B	B	A
6	B	A	B	A	26	B	B	A	B
7	B	$y=-2x+20$	$y=25x$	$y=15x+40$	27	B	A	A	A
8	$y=3x+6$	B	B	B	28	D	C	C	D
9	B	C	C	A	29	C	B	C	B
10	A	63	40	72	30	C	C	C	C
11	B	B	A	A	31	B	B	78.5	A
12	36	6	5	8	32	D	C	D	C
13	C	A	A	A	33	2	3	2	2
14	C	B	B	B	34	C	C	C	B
15	4	B	B	B	35	B	C	A	C
16	A	B	B	B	36	B	C	C	C
17	A	A	A	A	37	$160	$216	$300	$360
18	B	B	B	B	38	C	B	D	B
19	A,C	A,C	A,C	A,C	39	C	C	C	C
20	C	C	C	C	40	B	B	B	B

Quick Answer Keys — Reading Language Arts

#	T1	T2	T3	T4	#	T1	T2	T3	T4	#	T1	T2	T3	T4
1	A	B	B	B	11	B	B	B	B	21	B	B	B	B
2	B	B	B	B	12	B	B	A	B	22	C	B	B	B
3	C	C	B	B	13	C	B	B	B	23	B	B	B	B
4	B	A	B	B	14	B	B	B	B	24	A	A	A	A
5	B	B	B	A	SCR2	CR	CR	CR	CR	25	A	A	A	A

6	C	C	C	B	15	B	A	A	B	26	A	A	A	A
7	B	B	B	B	16	B	B	B	B	27	A	A	A	D
SCR1	CR	CR	CR	CR	17	B	B	B	B	28	A	A	A	A
8	B	B	B	B	18	B	B	B	B	ECR	CR	CR	CR	CR
9	B	B	B	B	19	B	A	B	B					
10	A	B	B	A	20	B	B	B	B					

The detailed explanations that follow show **why** each answer is correct — and why the common wrong choices are wrong. Read them even for questions you got right.

Mathematics — Answer Explanations

Each explanation gives the correct answer, a worked solution, the common wrong-answer trap, and the TEKS. ★ = Readiness, ○ = Supporting.

Practice Test 1

1. A *(8.2D ★)* — Convert to decimals: ⅔≈0.667, 68%=0.68, 0.7=0.70, √0.5≈0.707 → ⅔, 68%, 0.7, √0.5. *Trap: not evaluating √0.5.*

2. B *(8.2B ○)* — $6^2=36$, $7^2=49$; $36<40<49$ → between 6 and 7.

3. 6.8×10^4 *(8.2C ○)* — 68,000, decimal moved 4 places.

4. C *(8.2A ○)* — π is irrational; √36=6, ⅘, 0.25 are rational.

5. B *(8.4C ★)* — (11−2)/(4−1)=9/3=3. *Trap: D uses rise only.*

6. B *(8.4B ★)* — Unit rate = 12/3 = 4.

7. B *(8.4A ○)* — Similar right triangles → constant rise÷run.

8. $y = 3x + 6$ *(8.5I ★)* — Slope 3, y-intercept 6.

9. B *(8.5G ★)* — Each input one output; A/C/D repeat an x.

10. A *(8.5A ○)* — y=4x passes through the origin; others add a constant.

11. B *(8.5B ○)* — y = 6x + 15. *Trap: A swaps rate and fee.*

12. 36 *(8.5E ○)* — k=20/5=4; 4×9=36.

13. C *(8.5F ○)* — Ratios 4, 3.5, 3.33 not constant.

14. C *(8.5H ○)* — $2/lb, no fee → proportional.

15. 4 *(8.8C ★)* — 4x=16 → x=4.

16. A *(8.8A ○)* — 2x−3 = x+5.

17. A *(8.8B ○)* — 5x+10 = 3x+30.

18. B *(8.9A ○)* — Intersection (2,5): x=2, y=5.

19. A, C *(8.5I ★)* — Slope 2, y-intercept −3.

20. C *(8.7A ★)* — $\pi r^2h = 3.14\times16\times7 = 351.68$.

21. B *(8.7A ★)* — $\frac{1}{3}\pi r^2h = \frac{1}{3}\times3.14\times36\times10 = 376.8$. *Trap: C forgets ⅓.*

22. C *(8.7B ★)* — $48\pi+18\pi = 66\pi \approx 207.24$.

23. 25 *(8.7C ★)* — $49+576 = 625 = 25^2$.

24. A *(8.7D ○)* — Legs 4, 3 → 5.

25. A *(8.3C ★)* — (3×−2, 3×5) = (−6, 15).

26. B *(8.10C ★)* — (x,y)→(−x,y) → (−5, −4).

27. B *(8.10C ★)* — 90° CCW (x,y)→(−y,x) → (−7, 2).

28. D *(8.10A/B ○)* — Dilation changes size; the rest keep congruence.

29. C *(8.8D ○)* — 6x=180, x=30; largest 3x=90°.

30. C *(8.3A ○)* — 6×(5/2)=15.

31. B *(8.6A ○)* — B = base area = πr^2.

32. D *(8.10D ○)* — Area ×k^2: 16×9=144. *Trap: B uses ×3.*

33. 2 *(8.11B ○)* — Mean 6; MAD = (3+1+1+3)/4 = 2.

34. C *(8.5D ★)* — 1.5×8+2 = 14.

35. B *(8.11A ○)* — Falling points → negative linear.

36. B *(8.5C ○)* — Straight upward path → linear.

37. $160 *(8.12D ★)* — 800×0.05×4 = 160.

38. C *(8.12A ○)* — Longer loan → most interest.

39. C *(8.12C ○)* — Regular + compound → grows.

40. B *(8.12G ○)* — (12,000−4,000)/16 = 500.

Practice Test 2

1. A *(8.2D ★)* — ¾=0.75, 0.72, √0.6≈0.775, 78%=0.78 → 0.72, ¾, √0.6, 78%.

2. B *(8.2B ○)* — $81<90<100$ → 9 and 10.

3. 4.5×10^{-4} *(8.2C ○)* — 0.00045, 4 places.

4. B *(8.2A ○)* — √16=4 rational; others irrational.

5. B *(8.4C ★)* — 8/4 = 2.

6. A *(8.4B ★)* — 15/5 = 3.

7. $y = -2x + 20$ *(8.5I ★)* — Start 20, rate −2.

8. B *(8.5G ★)* — Each input once.

9. C *(8.5B ○)* — +4 → non-proportional.

10. 63 *(8.5E ○)* — k=9; 9×7=63.

11. B *(8.5F ○)* — Ratio 4 constant.

12. 6 *(8.8C ★)* — 4x=24 → x=6.

13. A *(8.8A ○)* — 3x+5 = 2x+12.

14. B *(8.9A ○)* — (−3,2): x=−3, y=2.

15. B *(8.4A ○)* — Similar triangles → same slope.

16. B *(8.5H ○)* — Steady 6 mph, no fee → proportional.

17. A *(8.8B ○)* — 4x+6 = 2x+18.

18. B *(8.5B ○)* — y = 3x + 4.

19. A, C *(8.5I ★)* — Slope −3, y-intercept 6.

20. C *(8.7A ★)* — 3.14×25×6 = 471.

21. C *(8.7A ★)* — 4⁄3×3.14×27 = 113.04.

22. B *(8.7B ★)* — 40π+8π = 48π ≈ 150.72.

23. 41 *(8.7C ★)* — 81+1600 = 1681 = 41².

24. B *(8.7D ○)* — √(36+64) = 10.

25. B *(8.3C ★)* — (8, −6).

26. B *(8.10C ★)* — (x,y)→(x,−y) → (−6, −2).

27. A *(8.10C ★)* — 180° → (−4, −1).

28. C *(8.10A/B ○)* — Translation preserves congruence.

29. B *(8.8D ○)* — Corresponding angles equal → 65°.

30. C *(8.3A ○)* — 9×(4/3)=12.

31. B *(8.6C ○)* — a²+b²=c².

32. C *(8.10D ○)* — Area ×k²: 15×4=60.

33. 3 *(8.11B ○)* — Mean 9; MAD = (5+1+1+5)/4 = 3.

34. C *(8.5D ★)* — 3×6+5 = 23.

35. C *(8.11A ○)* — No pattern → no association.

36. C *(8.5C ○)* — U-shape → non-linear.

37. $216 *(8.12D ★)* — 1,200×0.06×3 = 216.

38. B *(8.12A ○)* — Higher rate raises cost.

39. C *(8.12C ○)* — Regular + compound grows most.

40. B *(8.12G ○)* — (16,000−6,000)/20 = 500.

Practice Test 3

1. A *(8.2D ★)* — 0.6, 62%=0.62, ⅝=0.625, √0.4≈0.632.

2. B *(8.2B ○)* — 100<115<121 → 10 and 11.

3. 5.2×10⁶ *(8.2C ○)* — 5,200,000.

4. A *(8.2A ○)* — √45 irrational; √49=7.

5. A *(8.4C ★)* — 8/4 = 2.

6. B *(8.4B ★)* — 10/4 = 2.5.

7. y = 25x *(8.5I ★)* — 25 L/min, through origin.

8. B *(8.5G ★)* — Each input once.

9. C *(8.5B ○)* — y=2x−5 non-proportional.

10. 40 *(8.5E ○)* — k=8; 8×5=40.

11. A *(8.5F ○)* — Ratio 3 constant.

12. 5 *(8.8C ★)* — 5x=25 → x=5.

13. A *(8.8A ○)* — 4x−7 = x+8.

14. B *(8.9A ○)* — (4,−1).

15. B *(8.4A ○)* — Similar triangles.

16. B *(8.5H ○)* — Steady rate, no fee.

17. A *(8.8B ○)* — 6x+2 = 2x+22.

18. B *(8.5B ○)* — y=2x+5.

19. A, C *(8.5I ★)* — Slope 4, y-int −7.

20. C *(8.7A ★)* — 3.14×9×10 = 282.6.

21. B *(8.7A ★)* — ⅓×3.14×9×7 = 65.94.

22. C *(8.7B ★)* — 40π+32π = 72π ≈ 226.08.

23. 17 *(8.7C ★)* — 64+225 = 289.

24. A *(8.7D ○)* — Legs 4, 3 → 5.

25. B *(8.3C ★)* — (6, −10).

26. A *(8.10C ★)* — Over y-axis → (2, −7).

27. A *(8.10C ★)* — 180° → (3, −4).

28. C *(8.10D ○)* — Area ×k²: 10×9 = 90.

29. C *(8.8D ○)* — 40+75 = 115°.

30. C *(8.3A ○)* — 15×(8/5) = 24.

31. 78.5 *(8.6A ○)* — πr² = 3.14×25.

32. D *(8.10A/B ○)* — Dilation changes size.

33. 2 *(8.11B ○)* — Mean 8; MAD 2.

34. C *(8.5D ★)* — 2×12+1 = 25.

35. A *(8.11A ○)* — Rising → positive linear.

36. C *(8.5C ○)* — Straight rising → linear.

37. $300 *(8.12D ★)* — 1,500×0.04×5.

38. D *(8.12A ○)* — Longest loan → most interest.

39. C *(8.12C ○)* — Regular + compound grows.

40. B *(8.12G ○)* — (20,000−8,000)/24 = 500.

Practice Test 4

1. A *(8.2D ★)* — ⅞=0.875, 0.88, 89%=0.89, √0.8≈0.894.

2. B *(8.2B ○)* — 144<150<169 → 12 and 13.

3. 7.2×10^{-5} *(8.2C ○)* — 0.000072.

4. A *(8.2A ○)* — √24 irrational; √25=5.

5. B *(8.4C ★)* — −8/4 = −2.

6. A *(8.4B ★)* — 9/6 = 1.5.

7. y = 15x + 40 *(8.5I ★)* — Start 40, rate 15.

8. B *(8.5G ★)* — Each input once.

9. A *(8.5B ○)* — y=5x+2 non-proportional.

10. 72 *(8.5E ○)* — k=9; 9×8=72.

11. A *(8.5F ○)* — Ratio 5 constant.

12. 8 *(8.8C ★)* — 3x=24 → x=8.

13. A *(8.8A ○)* — 2x+9 = 5x−6.

14. B *(8.9A ○)* — (−2,−5).

15. B *(8.4A ○)* — Similar triangles.

16. B *(8.5H ○)* — $1.50 per 3 apples, steady.

17. A *(8.8B ○)* — 7x+5 = 3x+25.

18. B *(8.5B ○)* — y=0.05x+10.

19. A, C *(8.5I ★)* — Slope −2, y-int 8.

20. C *(8.7A ★)* — 3.14×36×5 = 565.2.

21. C *(8.7A ★)* — 4⁄3×3.14×216 = 904.32.

22. C *(8.7B ★)* — 40π+50π = 90π ≈ 282.6.

23. 29 *(8.7C ★)* — 400+441 = 841.

24. C *(8.7D ○)* — Legs 6, 8 → 10.

25. A *(8.3C ★)* — ½ scale → (−2, 3).

26. B *(8.10C ★)* — Over x-axis → (3, 8).

27. A *(8.10C ★)* — 90° CCW → (−2, 5).

28. D *(8.8D ○)* — 9x=180, x=20; largest 80°.

29. B *(8.10D ○)* — Area ×¼: 24→6.

30. C *(8.3A ○)* — 12×(7/4) = 21.

31. A *(8.6C ○)* — Hypotenuse² = sum of legs².

32. C *(8.10A/B ○)* — Rotation preserves congruence.

33. 2 *(8.11B ○)* — Mean 10; MAD 2.

34. B *(8.5D ★)* — 0.5×20+3 = 13.

35. C *(8.11A ○)* — No pattern → no association.

36. C *(8.5C ○)* — Curved → non-linear.

37. $360 *(8.12D ★)* — 2,000×0.03×6.

38. B *(8.12D ★)* — Compound earns more than simple.

39. C *(8.12C ○)* — Regular + compound grows most.

40. B *(8.12G ○)* — (24,000−9,000)/30 = 500.

Reading Language Arts — Answer Explanations

Each explanation gives the correct answer, the reasoning and text evidence, the common trap, and the TEKS. ★ = Readiness, ○ = Supporting. Constructed responses are scored with the rubrics in the SCR/ECR scored-samples section.

Practice Test 1

1. A *(2B ★)* — *Hovered* = rested lightly; fingers poised over the keys.

2. B *(7B ★)* — ¶3 recalls freezing at the talent show.

3. C *(5F ★)* — "playing anyway" = acting despite fear.

4. B *(7A ○)* — Facing fear leads to growth (theme).

5. B *(9F ○)* — She leaves "lighter" and "smiling" — calm, hopeful.

6. C *(6C ★)* — "played anyway... made all the difference" shows she overcame fear.

7. B *(7C ★)* — ¶3 jumps to an earlier event — a flashback.

SCR 1 *(6C/6D ★)* — Fearful → confident, with evidence.

8. B *(5G/8D ★)* — Central idea: urban beekeeping benefits cities.

9. B *(9A ★)* — Purpose: inform about the benefits.

10. A *(9B ○)* — ¶3 compares urban and rural hives.

11. B *(2B ★)* — Pollinators "move pollen from flower to flower."

12. B *(6C ★)* — "varied diet" supports healthier bees.

13. C *(5F ★)* — Avoiding pesticides → pesticides can harm bees.

14. B *(6D ★)* — Best summary: growth, benefits, training.

SCR 2 *(6C ★)* — One benefit + evidence + explanation.

15. B *(8B/9D ○)* — "red as stoplights" (as) → simile.

16. B *(9D ○)* — "city tilts its face" → personification.

17. B *(5G ★)* — Text B lists several benefits.

18. B *(5H ★)* — Both: gardens bring nature to the city.

19. B *(6B ★)* — Rainwater/cooling only in Text B.

20. B *(9A ★)* — The poem expresses a feeling.

21. B *(10C ★)* — 400-bottles detail develops the idea.

22. C *(10B·i ★)* — "In addition" adds a reason.

23. B *(10C ★)* — One clear sentence; avoids a comma splice.

24. A *(10B ★)* — Sentence 1 is the thesis.

25. A *(10D ○)* — "and me" → "and I."

26. A *(10D·i ★)* — *exhibit* singular → "was."

27. A *(10D·ii ★)* — "seen" → "saw."

28. A *(10D·vii ★)* — "Its" → "It's" (it is).

ECR *(11C/6B ★)* — Rubric: claim, organization, evidence, conventions.

Practice Test 2

1. B *(8C ○)* — Parentheses = stage directions.

2. B *(7B ★)* — Jamal jokes to lighten her nerves.

3. C *(5F ★)* — Pacing + "I can't do this" = nervous.

4. A *(7A ○)* — A friend helps her face fear (theme).

5. B *(2B ★)* — "Break a leg" = good luck (idiom).

6. C *(6C ★)* — "takes a deep breath... Okay" = courage.

7. B *(8C ○)* — The cue moves action toward her entrance.

SCR 1 *(6C ★)* — Jamal reassures her ("you're ready") + evidence.

8. B *(8E ★)* — Claim: phones should be put away.

9. B *(8E ★)* — ¶4 raises the opposing view to answer it.

10. B *(9A ★)* — Purpose: persuade.

11. B *(6C ★)* — "dozens of times an hour" = distraction.

12. B *(8E ○)* — Audience: leaders, students, parents.

13. B *(9G ○)* — Reasons + a counter-argument.

14. B *(6D ★)* — Summary states claim and reasons.

SCR 2 *(6C ★)* — Phones distract/harm + evidence.

15. A *(5G ★)* — Text A: teens need enough sleep.

16. B *(5G ★)* — Later start times → several benefits.

17. B *(5H ★)* — Both link sleep to school success.

18. B *(6B ★)* — Attendance/grades only in Text B.

19. A *(9B ○)* — Cause (body clock) and effects.

20. B *(9A ★)* — Both inform about teen sleep.

21. B *(10C ★)* — "study how plants grow" develops it.

22. B *(10B·i ★)* — "In addition" adds a reason.

23. B *(10C ★)* — One combined sentence is clearest.

24. A *(10B ★)* — Sentence 1 is the thesis.

25. A *(10D·i ★)* — "class go" → "class goes."

26. A *(10D ○)* — "me and my friends" → "my friends and I."

27. A *(10D·vii ★)* — "It's arms" → "Its arms."

28. A *(10D·vii ★)* — "alot" → "a lot."

ECR *(11B/6B ★)* — Rubric: controlling idea, organization, both-text evidence, conventions.

Practice Test 3

1. B *(2B ★)* — "nothing but boredom" = impatient/bored.

2. B *(7B ★)* — "I kept casting, because he kept casting."

3. B *(5F ★)* — "paying attention" = focused/learning.

4. B *(7A ○)* — Patience is valuable (theme).

5. B *(9A ★)* — Shares a personal experience and its lesson.

6. C *(6C ★)* — "The next cast, I waited... lifted gently."

7. B *(9D ○)* — "patience... its own kind of catch" → metaphor.

SCR 1 *(6C/6D ★)* — Learns patience, with evidence.

8. B *(9A ★)* — Purpose: persuade to support the library.

9. B *(8E ★)* — Message: act to save the library.

10. B *(9G ○)* — "Is that the school we want?" → rhetorical question.

11. B *(6C ★)* — "books they could never afford to buy."

12. A *(9G ○)* — "Stand up for your library" = call to action.

13. B *(5F ★)* — The urgency implies the library is at risk.

14. B *(6D ★)* — Summary: call to action + who depends on it.

SCR 2 *(6C ★)* — A reason it matters + evidence.

15. A *(9D ○)* — "patient, giving" tree → personification.

16. B *(8B ○)* — Lines and a stanza with rhyme.

17. B *(5G ★)* — Text B: trees provide benefits.

18. B *(5H ★)* — Both: trees are valuable, worth protecting.

19. B *(6B ★)* — Air/floods only in Text B.

20. B *(9A ★)* — Poem expresses appreciation.

21. B *(10C ★)* — "tools and clothing" detail develops it.

22. B *(10B·i ★)* — "In addition" adds a reason.

23. B *(10C ★)* — One combined sentence.

24. A *(10B ★)* — Sentence 1 is the thesis.

25. A *(10D·i ★)* — "Rex run" → "runs."

26. A *(10D·i ★)* — "He wait" → "waits."

27. A *(10D·vii ★)* — "Their is" → "There is."

28. A *(10D·vii ★)* — "are walk" → "our walk."

ECR *(11C/6B ★)* — Rubric-scored.

Practice Test 4

1. B *(2B ★)* — *Scavenged* = collected from discarded things.

2. B *(7B ★)* — She "promised to keep listening" to her father.

3. B *(5F ★)* — A repeating pattern → deliberate message.

4. B *(7A ○)* — Persistence leads to discovery (theme).

5. A *(9F ○)* — "never feel empty again" → hopeful, wonder.

6. B *(6C ★)* — "She had nearly given up."

7. B *(8A ○)* — A home-built radio + space signal → science fiction.

SCR 1 *(6C ★)* — Sky now feels full of possibility + evidence.

8. B *(5G ★)* — Fungi play a hidden, vital role.

9. B *(2B ★)* — Mycelium = web of tiny threads in soil.

10. A *(9B ○)* — Function (recycling) and its result.

11. B *(9A ★)* — Purpose: inform about fungi.

12. B *(6C ★)* — "connect to the roots of trees, trading nutrients."

13. B *(5F ★)* — Without fungi, dead material would pile up.

14. B *(6D ★)* — Summary: network + recycling.

SCR 2 *(6C ★)* — One way fungi help + evidence.

15. B *(8E ★)* — Claim: build more protected bike lanes.

16. B *(5G ★)* — Text B provides supporting data.

17. B *(5H ★)* — Both: bike lanes benefit safety and communities.

18. B *(6B ★)* — "40% fewer injuries" only in Text B.

19. B *(9A ★)* — Text A argues for bike lanes.

20. B *(9C ○)* — Text B adds facts/numbers as evidence.

21. B *(10C ★)* — "think several moves ahead" develops it.

22. B *(10B·i ★)* — "In addition" adds a reason.

23. B *(10C ★)* — One combined sentence.

24. A *(10B ★)* — Sentence 1 is the thesis.

25. A *(10D ○)* — "and me" → "and I."

26. A *(10D·i ★)* — *band* singular → "was."

27. D *(10D ○)* — Sentence 3 is already correct — no change.

28. A *(10D·vii ★)* — "hole" → "whole."

ECR *(11B ★)* — Rubric-scored.

Short Constructed Response — Scored Samples

Real-style responses on the **2-point** rubric. Remember: a 2-point answer = **Answer + Evidence + Explanation.**

Prompt (RLA Test 1, SCR 1)

How does Maya change from the beginning to the end of the passage? Use evidence from the text.

Score 2 — Full credit. "At the start, Maya is afraid and almost quits—she 'almost reached for her bag, ready to leave.' By the end, she has faced her fear and feels proud: 'She was smiling,' and she 'played anyway.' Maya changes from fearful to confident." *— clear answer, two pieces of evidence, explanation.*

Score 1 — Partial. "Maya was scared, but then she played the piano and felt better at the end." *— correct idea, but no text evidence.*

Score 0 — No credit. "I think Maya is a good piano player, and I like music too." *— does not answer the question; no evidence.*

Prompt (RLA Test 4, SCR 2)

Explain one way fungi help a forest. Use evidence from the passage.

Score 2 — Full credit. "Fungi help a forest by feeding trees. The text says the threads 'connect to the roots of trees, trading nutrients for sugar.' This means fungi give trees nutrients they need to grow." *— answer + quotation + explanation.*

Score 1 — Partial. "Fungi help trees grow in the forest." *— correct point, but no evidence.*

Score 0 — No credit. "Mushrooms can be food for people." *— off-topic.*

> **Self-check for every SCR:** Did I (1) answer the exact question, (2) include a specific detail or quotation, and (3) explain how it supports my answer? All three yes → a 2-point response.

Extended Constructed Response — Scored Samples

One ECR prompt answered at three levels. The **10-point** rubric rewards a clear claim, coherent organization, specific explained evidence, and correct conventions.

Prompt (RLA Test 1, ECR)

Should your community start an urban beekeeping or rooftop-garden program? Write an argumentative essay using evidence from "Cities Buzz Again."

Score 8–10 — High

> Cities should start urban beekeeping and rooftop-garden programs because they make communities healthier, greener, and more connected.
>
> First, bees and gardens improve the environment. According to "Cities Buzz Again," a single hive "can pollinate millions of flowers in a day," leading to "healthier parks, gardens, and street trees." Greener neighborhoods mean cleaner air for everyone.
>
> Second, these programs bring people together. The passage explains that "neighbors who tend a shared hive learn about nature and work together," and local honey "supports small businesses." A community that grows together grows stronger.
>
> Some people worry that beekeeping is dangerous or difficult. It is true that "beekeeping takes training." However, with proper instruction, hives are safe, and schools already use them to teach science. The benefits clearly outweigh the risks.
>
> For these reasons, my community should start an urban beekeeping or rooftop-garden program. It would make our city healthier, greener, and more united.

Why it scores high: clear claim; two organized body paragraphs with explained passage evidence; a counter-argument that is answered; a strong conclusion; correct conventions.

Score 4–7 — Middle

"I think cities should have bees because they help plants. The article says bees pollinate flowers, and that is good for gardens. Also, honey is nice to have. Some people don't like bees, but they are okay. Cities should try beekeeping." *— a position and one piece of evidence, but thin development, weak organization, vague counter-argument.*

Score 1–3 — Low

"Bees are good. I like gardens. Cities should do it because honey is sweet." *— a position but no real evidence, no organization, little development.*

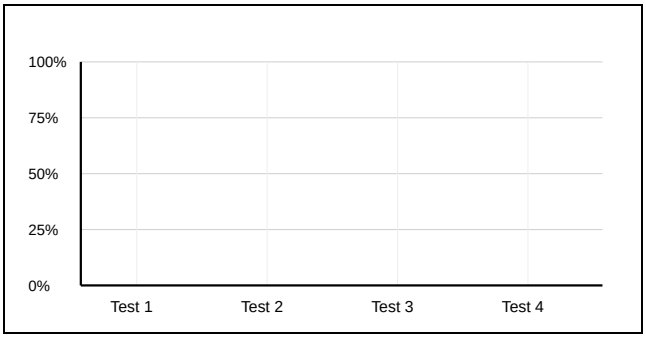
Planning pays off. Jot your claim and two reasons with evidence before writing. An organized essay with explained evidence almost always scores higher than a longer, unplanned one. Leave a minute to check spelling and punctuation.

Part 4 · Diagnose & Grow

Scoring a test is only step one. The real gains come from seeing **which skills need work** and studying those next. Use these pages after every practice test.

Score-Tracking Sheet

Plot your scores and watch them climb



After each test, put a dot at your percent, then connect the dots.

After each test, count the questions you answered correctly and write it here. Watch your scores climb from Test 1 to Test 4.

Test	Date	Points Earned	Out Of	Estimated Level
Math 1			~48	
Math 2			~48	
Math 3			~48	
Math 4			~48	
RLA 1			~56	
RLA 2			~56	
RLA 3			~56	
RLA 4			~56	

If you earned about…	Estimated level
Less than ~35%	Did Not Meet Grade Level
~35%–55%	Approaches Grade Level

~55%–80%	Meets Grade Level
~80% or more	Masters Grade Level

Important: The State of Texas sets the official passing scores each year, and they change. These percentages are a **rough guide** to track progress, not an official score. Aim for **Meets** or higher.

Strand Diagnostic Chart

For each test, count how many you got right in each strand or section. A low percentage points you straight to the lesson to review.

Mathematics strand	Qs / test	If low, review…
1 · Numerical Representations & Relationships	~4	Strand 1 lesson
2 · Computations & Algebraic Relationships	~15	Strand 2 lesson
3 · Geometry & Measurement	~13	Strand 3 lesson
4 · Data Analysis & Personal Financial Literacy	~8	Strand 4 lesson

RLA skill area	If low, review…
Vocabulary (context clues, roots)	Reading Lessons, Part A
Comprehension & inference	Reading Lessons, Part B
Using text evidence / summarizing	Reading Lessons, Part C
Literary elements	Reading Lessons, Part D
Author's purpose & craft	Reading Lessons, Part E
Revising & Editing	Revising & Editing Refresher
Short / Extended responses	SCR & ECR Mastery

How to use it: After each test, find your **weakest** row, reread that lesson, redo its "Try It" items, then try the next practice test. This is the fastest way to raise your score.

A Guide for Parents and Teachers

This book is built so a student can use it alone, but your support makes it far more powerful.

Start With the Diagnostics

The **Score-Tracking Sheet** and **Strand Diagnostic Chart** turn each test into a study plan. After a test, help the student find the weakest strand or skill area and go back to that lesson first. Watch for scores rising across Tests 1–4.

Coaching the Written Responses

The SCR and ECR are where many students leave points on the table. You do not need to grade like the state; instead, ask three questions:

1. **Did you answer the exact question?**
2. **Did you use specific evidence from the text?**
3. **Did you explain how the evidence supports your point?**

Compare the student's writing to the **scored samples** in this book. Seeing the difference between a high and a middle response teaches more than a grade ever could.

Keep It Positive and Steady

- Short, regular practice beats long, rare cram sessions. Use the 4-week or 8-week calendar.
- Praise **effort and strategy** ("You used two pieces of evidence!"), not just the score.
- Normalize mistakes—every wrong answer with a good explanation is a lesson learned.

On Test Day

Remind the student of the **Test-Day Game Plan**: read directions carefully, answer easy questions first, never leave a blank, and give the written responses real effort.

> **A note on the 2027 test:** Spring 2027 is the final STAAR administration for grades 3–8. Whatever assessment follows, the skills in this book—reading closely, reasoning with evidence, solving real-world math—will continue to serve your student well.

TEKS Coverage Index

Every eligible Grade 8 Math and RLA student expectation, and where each is **taught** and **practiced**. ★ = Readiness, ○ = Supporting. Nothing outside these standards appears in the book, and nothing is left out.

Mathematics

TEKS	Skill	R/S	Taught	Practiced
8.2A	Sets of real numbers	○	Strand 1	Tests 1–4, Drills
8.2B	Approximate irrationals	○	Strand 1	Tests 1–4, Drills
8.2C	Scientific notation	○	Strand 1	Tests 1–4, Drills
8.2D	Order real numbers	★	Strand 1	Tests 1–4, Drills
8.4A	Slope via similar triangles	○	Strand 2	Tests 1–4
8.4B	Graph proportional; unit rate	★	Strand 2	Tests 1–4
8.4C	Rate of change & y-intercept	★	Strand 2	Tests 1–4, Drills
8.5A/B/E/F/H	Proportional / non-proportional, functions	○	Strand 2	Tests 1–4, Drills
8.5G	Identify functions	★	Strand 2	Tests 1–4, Drills
8.5I	Write y = mx + b	★	Strand 2	Tests 1–4, Drills
8.8A/B	Write equations / problems	○	Strand 2	Tests 1–4
8.8C	Solve equations, both sides	★	Strand 2	Tests 1–4, Drills
8.9A	Simultaneous linear (intersection)	○	Strand 2	Tests 1–4
8.3A/B	Similar shapes, dilations	○	Strand 3	Tests 1–4
8.3C	Dilation by scale factor	★	Strand 3	Tests 1–4, Drills
8.6A/C	Cylinder volume; Pythagorean model	○	Strand 3	Tests 1–4
8.7A	Volume: cylinders, cones, spheres	★	Strand 3	Tests 1–4, Drills
8.7B	Surface area	★	Strand 3	Tests 1–4
8.7C	Pythagorean theorem & converse	★	Strand 3	Tests 1–4, Drills
8.7D	Distance between two points	○	Strand 3	Tests 1–4
8.8D	Angle relationships, AA	○	Strand 3	Tests 1–4, Drills
8.10A/B	Transformations & congruence	○	Strand 3	Tests 1–4
8.10C	Translations, reflections, rotations	★	Strand 3	Tests 1–4, Drills
8.10D	Dilation effect on measurement	○	Strand 3	Tests 1–4
8.5C	Bivariate data: linear vs. not	○	Strand 4	Tests 1–4
8.5D	Trend line predictions	★	Strand 4	Tests 1–4, Drills
8.11A	Scatterplot & association	○	Strand 4	Tests 1–4, Drills
8.11B	Mean absolute deviation	○	Strand 4	Tests 1–4, Drills
8.12A/C/G	Cost of credit, saving, college plan	○	Strand 4	Tests 1–4
8.12D	Simple vs. compound interest	★	Strand 4	Tests 1–4, Drills

8.1 A–G	Mathematical process standards	—	Process Standards	Woven through all items

Reading Language Arts

TEKS	Skill	R/S	Taught	Practiced
2A/2C	Reference tools; Greek/Latin roots	○	Reading A	RLA Tests 1–4
2B	Context clues	★	Reading A	RLA Tests 1–4
5C/5E	Predictions; connections	○/★	Reading B	Reading lessons
5F	Inference + evidence	★	Reading B	RLA Tests 1–4
5G	Key ideas / central idea	★	Reading B	RLA Tests 1–4
5H	Synthesize	★	Reading B	RLA Tests 1–4
6B	Compare sources	★	ECR Mastery	Paired sets + ECR
6C	Text evidence	★	Reading C	Tests 1–4, SCR
6D	Paraphrase & summarize	★	Reading C	RLA Tests 1–4
7A/7D	Theme; setting	○	Reading D	RLA Tests 1–4
7B	Character motivation	★	Reading D	RLA Tests 1–4
7C	Non-linear plot	★	Reading D	RLA Test 1
8A	Literary genres	○	Genre Guide	RLA Test 4
8B	Poetry: graphical elements	○	Genre Guide	RLA Tests 1, 3
8C	Drama: acts/scenes	○	Genre Guide	RLA Test 2
8D	Informational: thesis, features	★/○	Genre Guide	RLA Tests 1–4
8E	Argumentative: claim, counter, audience	★/○	Genre Guide	RLA Tests 2, 4
9A	Author's purpose & message	★	Reading E	RLA Tests 1–4
9B/9C	Text structure; print/graphic features	○	Reading E	RLA Tests 1–4
9D/9E/9F/9G	Figurative language, devices, tone, rhetoric	○	Reading E	RLA Tests 1–4
10B/10C	Composition: structure, revise	★	Revising/Editing	RLA Tests 1–4
10D·i/ii/vii	Sentences, verb tense, spelling	★	Revising/Editing	RLA Tests 1–4
10D·iii–vi	Prepositions, pronoun, caps, punctuation	○	Revising/Editing	RLA Tests 1–4
11B	Compose informational	★	ECR Mastery	RLA Tests 2, 4 (ECR)
11C	Compose argumentative	★	ECR Mastery	RLA Tests 1, 3 (ECR)
11D	Compose correspondence	○	ECR Mastery	Revising/Editing

Coverage guarantee: Every eligible Grade 8 Math and RLA TEKS is both taught and practiced in this book, and no content outside the eligible TEKS is included.

Reading & Writing Warm-Up Drills

Quick, focused drills to sharpen one skill at a time before a full-length test. Each item names the TEKS it practices. Answers are at the end.

Vocabulary (2B, 2C)

1. In "The storm was *relentless*, never stopping for days," *relentless* means — A. gentle B. unending C. short D. quiet
2. The root *path* (feeling) shows that *sympathy* means — A. a feeling with or for another B. a path C. a star D. a leader
3. The root *duc* means *lead*, so *conduct* most likely means — A. to follow B. to lead or guide C. to stop D. to break

Comprehension & Inference (5F, 5G)

4. A character "grabbed an umbrella and sighed at the gray sky." You can infer that — A. it is sunny B. it may rain C. she is joyful D. it is midnight
5. The sentence that states what a paragraph is mostly about is the — A. minor detail B. central idea C. quotation D. title

Author's Craft (9A, 9D, 9G)

6. "The wind *whispered* through the trees" is an example of — A. simile B. personification C. hyperbole D. irony
7. "As busy as a bee" is a — A. metaphor B. simile C. fact D. fallacy
8. An author who writes to convince you to recycle has the purpose to — A. entertain B. inform C. persuade D. confuse
9. "Everybody owns this phone, so you should too" is which fallacy? — A. bandwagon B. circular reasoning C. analogy D. irony

Literary Elements (7B, 7C)

10. A scene that jumps back to a character's childhood is a — A. flashback B. climax C. setting D. theme
11. An early hint about a later event is — A. irony B. foreshadowing C. mood D. plot

Revising & Editing (10B, 10C, 10D)

12. Which transition shows contrast? — A. In addition B. However C. For example D. First
13. The best sentence to add a detail is one that — A. repeats the topic B. gives a specific fact or example C. changes the subject D. asks a question
14. "The team of players *are* ready." Best change — A. *are → is* B. *team → teams* C. *players → player* D. no change
15. "*Its* going to rain." Best change — A. *Its → It's* B. *going → gone* C. *rain → Rain* D. no change

16. "*Me and him* went." Best change — A. *Me and him → He and I* B. *went → gone* C. no change D. add a comma

17. "*There* going to *their* house." The first word should be — A. *They're* B. *Their* C. no change D. deleted

18. "We *seen* the movie." Best change — A. *seen → saw* B. *movie → movies* C. no change D. *We → Us*

Answer Key

1. B — 2B · 2. A — 2C · 3. B — 2C · 4. B — 5F · 5. B — 5G · 6. B — 9D · 7. B — 9D · 8. C — 9A · 9. A — 9G · 10. A — 7C · 11. B — 7C · 12. B — 10B·i · 13. B — 10C · 14. A — 10D·i · 15. A — 10D·vii · 16. A — 10D · 17. A — 10D·vii · 18. A — 10D·ii

Glossary of Literary & Craft Terms

The terms the STAAR Grade 8 Reading standards expect you to know. Questions often use these exact words.

Story elements

Theme — the message or lesson, shown through characters and events. *(7A)*

Character motivation — why a character acts as they do. *(7B)*

Flashback — a scene that jumps back to an earlier time. *(7C)*

Foreshadowing — an early hint about what will happen. *(7C)*

Subplot — a smaller story beside the main plot. *(7C)*

Setting — when and where a story takes place. *(7D)*

Figurative language & devices

Simile — a comparison using *like/as*. *(9D)*

Metaphor — a comparison without *like/as*. *(9D)*

Extended metaphor — a metaphor developed over several lines. *(9D)*

Personification — human traits given to non-human things. *(9D)*

Irony — a contrast between expectation and reality. *(9E)*

Point of view — who tells the story. *(9E)*

Mood — the feeling a text creates. *(9F)*

Tone — the author's attitude. *(9F)*

Voice — the author's distinct personality. *(9F)*

Rhetoric & argument

Analogy — explaining one thing using another. *(9G)*

Juxtaposition — placing two things side by side for contrast. *(9G)*

Bandwagon appeal — "everyone's doing it." *(9G)*

Circular reasoning — using a claim to prove itself. *(9G)*

Claim — the position an argument defends. *(8E)*

Counter-argument — the opposing side, answered. *(8E)*

Audience — the readers a text is written for. *(8E)*

Thesis / controlling idea — the main point. *(8D, 10B)*

Genres

Poetry — lines and stanzas; watch line length and punctuation. *(8B)*

Drama — a play in acts and scenes with stage directions. *(8C)*

Informational / argumentative / persuasive — explains / defends a claim / convinces.

Correspondence — letters and emails. *(11D)*

Grammar & Conventions Quick Reference

Everything the Grade 8 editing questions check, on one page. (Standard 10D.)

Sentences (10D·i)

- **Complete sentence:** a subject + a verb + a complete thought.
- **Fragment:** "Running to the bus." → "She was running to the bus."
- **Run-on / comma splice:** "I was tired, I slept." → "I was tired, so I slept."

Subject-Verb Agreement (10D·i)

The verb matches the **subject**, not a word in between: "The **box** of markers **is** here" (ignore *of markers*). Singular → *is, has, runs*; plural → *are, have, run*.

Verb Tense & Voice (10D·ii)

Keep tense **consistent**. Use correct forms: *saw* (not "seen"), *gone* (not "went" after *have*).

Pronouns (10D·iii, iv)

Case: *I, he, she, we, they* as subjects; *me, him, her, us, them* as objects ("He and **I** went"; "between you and **me**").
Agreement: a pronoun matches its noun in number.

Capitalization & Punctuation (10D·v, vi)

- Capitalize sentence starts, proper nouns, and *I*.
- **Comma** around a nonrestrictive phrase: "My dog, a beagle, barks."
- **Semicolon** joins related sentences: "It rained; we stayed in."
- **Colon** introduces a list: "Bring three things: a pen, paper, and a book."
- **Parentheses** add extra info: "The test (held in April) was long."

Commonly Confused Words (10D·vii)

Word	Means	Example
its / it's	belonging to it / it is	The dog wagged its tail. It's raining.
there / their / they're	a place / belonging to them / they are	There it is. Their house. They're here.

to / two / too	toward / 2 / also	Go to class. Two pens. Me too.
affect / effect	to change / a result	Rain affects plants. The effect was clear.
your / you're	belonging to you / you are	Your book. You're right.
our / are	belonging to us / verb	Our class. We are ready.
whole / hole	entire / an opening	The whole day. A hole in the sock.
a lot	many (two words)	A lot of people came.

Greek & Latin Roots, Prefixes & Suffixes

Word parts help you unlock unfamiliar words. (Supports vocabulary standard 2C.)

Roots named in the Grade 8 standard	
ast	star — astronomy
qui	quiet, rest — tranquil
path	feeling — sympathy
mand / mend	order — command
duc	lead — conduct

More common roots	
port	carry — transport
spect	look — inspect
dict	say — predict
scrib/script	write — manuscript
tele	far — telescope
bio	life — biology
graph	write — paragraph
geo	earth — geography
aud	hear — audible
vis / vid	see — visible

Common prefixes	
re-	again, back — return
un- / in-	not — unfair, incorrect
pre-	before — preview
mis-	wrongly — mistake
dis-	opposite — disagree
inter-	between — interact
sub-	under — submarine

Common suffixes	
-able / -ible	able to be — readable
-tion / -sion	the act of — action
-ology	the study of — biology
-ist	one who — scientist
-ful	full of — helpful
-less	without — hopeless

Strategy: Break a hard word into parts. *Transport* = *trans* (across) + *port* (carry) → to carry across. Combine word-part clues with the **context** of the sentence (2B) for your best meaning.

Practice Answer Sheets

The real STAAR test is taken online, but marking your answers here while you practice makes scoring quick and keeps your work organized. For each multiple-choice question, fill in the letter of your answer. For **griddable, equation-editor, and "Select TWO" items**, write your answer in the box or circle both letters. Write your **SCR and ECR responses on separate lined paper.**

After the test, score with the **Quick Answer Keys**, then fill in your **Score-Tracking Sheet** and **Strand Diagnostic Chart.**

Mathematics · Practice Test 1 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**31.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** ☐	**22.** Ⓐ Ⓑ Ⓒ Ⓓ	**32.** Ⓐ Ⓑ Ⓒ Ⓓ
3. ☐	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** ☐	**33.** ☐
4. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ	**34.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**15.** ☐	**25.** Ⓐ Ⓑ Ⓒ Ⓓ	**35.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ	**36.** Ⓐ Ⓑ Ⓒ Ⓓ
7. Ⓐ Ⓑ Ⓒ Ⓓ	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ	**37.** ☐
8. ☐	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ	**38.** Ⓐ Ⓑ Ⓒ Ⓓ
9. Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**29.** Ⓐ Ⓑ Ⓒ Ⓓ	**39.** Ⓐ Ⓑ Ⓒ Ⓓ
10. Ⓐ Ⓑ Ⓒ Ⓓ	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**30.** Ⓐ Ⓑ Ⓒ Ⓓ	**40.** Ⓐ Ⓑ Ⓒ Ⓓ

Item 19 is "Select TWO" — circle both letters. Items with a box are griddable or equation-editor; write your answer in the box.

Mathematics · Practice Test 2 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**31.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** ☐	**22.** Ⓐ Ⓑ Ⓒ Ⓓ	**32.** Ⓐ Ⓑ Ⓒ Ⓓ
3. ☐	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** ☐	**33.** ☐
4. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ	**34.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ	**35.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ	**36.** Ⓐ Ⓑ Ⓒ Ⓓ
7. ☐	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ	**37.** ☐
8. Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ	**38.** Ⓐ Ⓑ Ⓒ Ⓓ
9. Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**29.** Ⓐ Ⓑ Ⓒ Ⓓ	**39.** Ⓐ Ⓑ Ⓒ Ⓓ
10. ☐	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**30.** Ⓐ Ⓑ Ⓒ Ⓓ	**40.** Ⓐ Ⓑ Ⓒ Ⓓ

Mathematics · Practice Test 3 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**31.** ☐
2. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** ☐	**22.** Ⓐ Ⓑ Ⓒ Ⓓ	**32.** Ⓐ Ⓑ Ⓒ Ⓓ
3. ☐	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** ☐	**33.** ☐
4. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ	**34.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ	**35.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ	**36.** Ⓐ Ⓑ Ⓒ Ⓓ
7. ☐	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ	**37.** ☐
8. Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ	**38.** Ⓐ Ⓑ Ⓒ Ⓓ
9. Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**29.** Ⓐ Ⓑ Ⓒ Ⓓ	**39.** Ⓐ Ⓑ Ⓒ Ⓓ
10. ☐	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**30.** Ⓐ Ⓑ Ⓒ Ⓓ	**40.** Ⓐ Ⓑ Ⓒ Ⓓ

Item 19 is "Select TWO" — circle both letters. Items with a box are griddable or equation-editor; write your answer in the box.

Mathematics · Practice Test 4 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**31.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** ☐	**22.** Ⓐ Ⓑ Ⓒ Ⓓ	**32.** Ⓐ Ⓑ Ⓒ Ⓓ
3. ☐	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** ☐	**33.** ☐
4. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ	**34.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ	**35.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ	**36.** Ⓐ Ⓑ Ⓒ Ⓓ
7. ☐	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ	**37.** ☐
8. Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ	**38.** Ⓐ Ⓑ Ⓒ Ⓓ
9. Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**29.** Ⓐ Ⓑ Ⓒ Ⓓ	**39.** Ⓐ Ⓑ Ⓒ Ⓓ
10. ☐	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**30.** Ⓐ Ⓑ Ⓒ Ⓓ	**40.** Ⓐ Ⓑ Ⓒ Ⓓ

Item 19 is "Select TWO" — circle both letters. Items with a box are griddable or equation-editor; write your answer in the box.

Reading Language Arts · Practice Test 1 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**8.** Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**22.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**9.** Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** Ⓐ Ⓑ Ⓒ Ⓓ
3. Ⓐ Ⓑ Ⓒ Ⓓ	**10.** Ⓐ Ⓑ Ⓒ Ⓓ	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ
4. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ
7. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ

Write SCR and ECR responses on separate lined paper. Items 1–28 are shown; SCR 1 follows item 7 and SCR 2 follows item 14 on the test.

Reading Language Arts · Practice Test 2 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**8.** Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**22.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**9.** Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** Ⓐ Ⓑ Ⓒ Ⓓ
3. Ⓐ Ⓑ Ⓒ Ⓓ	**10.** Ⓐ Ⓑ Ⓒ Ⓓ	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ
4. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ
7. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ

Reading Language Arts · Practice Test 3 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**8.** Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**22.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**9.** Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** Ⓐ Ⓑ Ⓒ Ⓓ
3. Ⓐ Ⓑ Ⓒ Ⓓ	**10.** Ⓐ Ⓑ Ⓒ Ⓓ	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ
4. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ
7. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ

Write SCR and ECR responses on separate lined paper. Items 1–28 are shown; SCR 1 follows item 7 and SCR 2 follows item 14 on the test.

Reading Language Arts · Practice Test 4 — Answer Sheet

1. Ⓐ Ⓑ Ⓒ Ⓓ	**8.** Ⓐ Ⓑ Ⓒ Ⓓ	**15.** Ⓐ Ⓑ Ⓒ Ⓓ	**22.** Ⓐ Ⓑ Ⓒ Ⓓ
2. Ⓐ Ⓑ Ⓒ Ⓓ	**9.** Ⓐ Ⓑ Ⓒ Ⓓ	**16.** Ⓐ Ⓑ Ⓒ Ⓓ	**23.** Ⓐ Ⓑ Ⓒ Ⓓ
3. Ⓐ Ⓑ Ⓒ Ⓓ	**10.** Ⓐ Ⓑ Ⓒ Ⓓ	**17.** Ⓐ Ⓑ Ⓒ Ⓓ	**24.** Ⓐ Ⓑ Ⓒ Ⓓ
4. Ⓐ Ⓑ Ⓒ Ⓓ	**11.** Ⓐ Ⓑ Ⓒ Ⓓ	**18.** Ⓐ Ⓑ Ⓒ Ⓓ	**25.** Ⓐ Ⓑ Ⓒ Ⓓ
5. Ⓐ Ⓑ Ⓒ Ⓓ	**12.** Ⓐ Ⓑ Ⓒ Ⓓ	**19.** Ⓐ Ⓑ Ⓒ Ⓓ	**26.** Ⓐ Ⓑ Ⓒ Ⓓ
6. Ⓐ Ⓑ Ⓒ Ⓓ	**13.** Ⓐ Ⓑ Ⓒ Ⓓ	**20.** Ⓐ Ⓑ Ⓒ Ⓓ	**27.** Ⓐ Ⓑ Ⓒ Ⓓ
7. Ⓐ Ⓑ Ⓒ Ⓓ	**14.** Ⓐ Ⓑ Ⓒ Ⓓ	**21.** Ⓐ Ⓑ Ⓒ Ⓓ	**28.** Ⓐ Ⓑ Ⓒ Ⓓ

Write SCR and ECR responses on separate lined paper. Items 1–28 are shown; SCR 1 follows item 7 and SCR 2 follows item 14 on the test.

Bonus · Mathematics Challenge Test

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

An extra full-length test for one more round of practice. Mark your answers on blank paper, then score with the key at the end.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: ⅞, 0.9, 91%, √0.8. A. ⅞, √0.8, 0.9, 91% B. 91%, 0.9, √0.8, ⅞ C. 0.9, ⅞, 91%, √0.8 D. √0.8, ⅞, 0.9, 91%

2. Between which two whole numbers does √60 lie? A. 6 and 7 B. 7 and 8 C. 29 and 31 D. 59 and 61

3. *(Griddable)* Write 83,000 in scientific notation. ☐

4. Which number is **irrational**? A. √30 B. √36 C. ⅗ D. 0.2

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (2, 5) and (5, 14)? A. 2 B. 3 C. 4 D. 9

6. A proportional relationship passes through the origin and (8, 20). What is the unit rate? A. 2 B. 2.5 C. 4 D. 12

7. *(Equation editor)* A fund starts at $100 and grows $8 each week. Write an equation for the amount y after x weeks. ☐

8. Which relation is a **function**? A. {(1, 4), (1, 6), (2, 7)} B. {(1, 4), (2, 4), (3, 7)} C. {(5, 1), (5, 2), (6, 3)} D. {(0, 1), (0, 2), (1, 3)}

9. Which equation represents a **non-proportional** relationship? A. $y = 7x$ B. $y = \frac{2}{5}x$ C. $y = 4x - 3$ D. $y = 9x$

10. *(Griddable)* For a direct variation, $y = 42$ when $x = 6$. What is y when $x = 9$? ☐

11. Which table shows a **proportional** relationship? A. (1, 4)(2, 8)(3, 12) B. (1, 3)(2, 7)(3, 11) C. (1, 5)(2, 8)(3, 11) D. (2, 9)(3, 13)(4, 17)

12. *(Griddable)* Solve for x: $10x - 4 = 6x + 16$. ☐

13. "Four less than three times a number equals the number plus six." Which equation models this? A. $3x - 4 = x + 6$ B. $4 - 3x = x + 6$ C. $3x + 4 = x - 6$ D. $4x - 3 = x + 6$

14. Two lines intersect at (5, 2). What is the solution of the system? A. $x = 2, y = 5$ B. $x = 5, y = 2$ C. $x = 7$ D. No solution

15. On a line, the right triangles drawn to any two points are — A. congruent B. **similar** C. equilateral D. obtuse

16. Which situation is **proportional**? A. $20 plus $15 per hour B. 5 oranges for $2 at a steady rate C. 40 L draining 5 L/min D. $8 fee plus $3 each

17. Which situation can be modeled by $5x + 8 = 3x + 20$? A. Two savers start with $8 and $20, saving $5 and $3 per week; when equal? B. 5 times 8 C. A rectangle's area D. 8 plus 20

18. A florist charges $6 plus $2.50 per flower. Which equation gives the cost y for x flowers? A. $y = 6x + 2.5$ B. $y = 2.5x + 6$ C. $y = 8.5x$ D. $y = 2.5x - 6$

19. *(Select TWO)* Which statements about $y = 5x - 2$ are true? A. The slope is 5 B. The slope is −2 C. The y-intercept is −2 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 2 and height 9. What is its volume? ($\pi \approx 3.14$) A. 56.52 B. 113.04 C. 226.08 D. 339.12

21. A cone has radius 6 and height 5. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 188.4 C. 376.8 D. 565.2

22. A cylinder has radius 3 and height 6. What is its **total** surface area? ($\pi \approx 3.14$) A. 113.04 B. 150.72 C. 169.56 D. 226.08

23. *(Griddable)* A right triangle has legs of 12 and 16. Find the hypotenuse. ☐

24. What is the distance between (−1, 2) and (2, 6)? A. 5 B. 7 C. 12 D. 25

25. The point (5, −2) is dilated by a scale factor of 3 about the origin. What is the image? A. (15, −6) B. (8, 1) C. (15, −2) D. (5, −6)

26. The point (−4, 9) is reflected over the x-axis. What is the image? A. (4, 9) B. (−4, −9) C. (4, −9) D. (9, −4)

27. The point (6, −3) is rotated 180° about the origin. What is the image? A. (−6, 3) B. (−3, 6) C. (3, −6) D. (6, 3)

28. An exterior angle of a triangle equals the sum of its two remote interior angles, 55° and 60°. What is the exterior angle? A. 65° B. 95° C. 115° D. 125°

29. Two similar figures have corresponding sides in a ratio of 2 : 3. A side of the smaller is 10. What is the matching side of the larger? A. 12 B. 13 C. 15 D. 20

30. A rectangle measures 8 by 2. It is dilated by a scale factor of 3. What is the area of the image? A. 16 B. 48 C. 96 D. 144

31. *(Griddable)* The base of a cylinder is a circle with radius 10. What is the base area B? ($\pi \approx 3.14$) ☐

32. Which transformation **changes** the size of a figure? A. Translation B. Reflection C. Rotation D. Dilation by factor 3

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 3, 3, 9, 9. ☐

34. A trend line is modeled by y = 2.5x + 4. Predict *y* when x = 8. A. 20 B. 22 C. 24 D. 28

35. On a scatterplot, the points fall from upper-left to lower-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which pattern suggests the data are **linear**? A. A U-shaped curve B. A circular cluster C. A straight rising or falling path D. Random scatter

37. *(Griddable)* Find the simple interest on $2,500 at 4% for 3 years. ☐

38. For the same loan amount and length, how does a **higher** interest rate affect the total cost? A. Lowers it B. Raises it C. No effect D. Cancels the loan

39. What helps savings grow the most over time? A. Spending quickly B. Saving once C. Regular saving with compound interest D. Keeping cash at home

40. College costs $18,000. A family contributes $6,000 and saves the rest over 24 months. How much per month? A. $400 B. $500 C. $600 D. $750

Bonus Test — Answer Key & Explanations

1. A *(8.2D ★)* — ⅞=0.875, √0.8≈0.894, 0.9, 91%=0.91.

2. B *(8.2B ○)* — 49<60<64.

3. 8.3×10⁴ *(8.2C ○)* — 83,000.

4. A *(8.2A ○)* — √30 irrational; √36=6.

5. B *(8.4C ★)* — 9/3=3.

6. B *(8.4B ★)* — 20/8=2.5.

7. y = 8x + 100 *(8.5I ★)* — Start 100, rate 8.

8. B *(8.5G ★)* — Each input once.

9. C *(8.5B ○)* — −3 → non-proportional.

10. 63 *(8.5E ○)* — k=7; 7×9.

11. A *(8.5F ○)* — Ratio 4 constant.

12. 5 *(8.8C ★)* — 4x=20.

13. A *(8.8A ○)* — 3x−4=x+6.

14. B *(8.9A ○)* — (5,2).

15. B *(8.4A ○)* — Similar triangles.

16. B *(8.5H ○)* — Steady rate, no fee.

17. A *(8.8B ○)* — 5x+8=3x+20.

18. B *(8.5B ○)* — y=2.5x+6.

19. A, C *(8.5I ★)* — Slope 5, y-int −2.

20. B *(8.7A ★)* — 3.14×4×9=113.04.

21. B *(8.7A ★)* — ⅓×3.14×36×5=188.4.

22. C *(8.7B ★)* — 36π+18π=54π≈169.56.

23. 20 *(8.7C ★)* — 144+256=400.

24. A *(8.7D ○)* — Legs 3,4→5.

25. A *(8.3C ★)* — (15,−6).

26. B *(8.10C ★)* — Over x-axis → (−4,−9).

27. A *(8.10C ★)* — 180° → (−6,3).

28. C *(8.8D ○)* — 55+60=115°.

29. C *(8.3A ○)* — 10×3/2=15.

30. D *(8.10D ○)* — Area ×9: 16×9=144.

31. 314 *(8.6A ○)* — 3.14×100.

32. D *(8.10A/B ○)* — Dilation changes size.

33. 3 *(8.11B ○)* — Mean 6; all distances 3.

34. C *(8.5D ★)* — 2.5×8+4=24.

35. B *(8.11A ○)* — Falling → negative linear.

36. C *(8.5C ○)* — Straight path → linear.

37. $300 *(8.12D ★)* — 2,500×0.04×3.

38. B *(8.12A ○)* — Higher rate raises cost.

39. C *(8.12C ○)* — Regular + compound grows most.

40. B *(8.12G ○)* — (18,000−6,000)/24=500.

Bonus · Reading Language Arts Challenge Test

Read each passage, then answer the questions that follow. Use evidence from the texts.

An extra reading test for one more round of practice. Mark answers on blank paper; write SCR and ECR responses on lined paper.

Passage 1 — The Last Lap (fiction)

(1) Diego's legs burned as he rounded the final turn. Three runners were ahead of him, and the finish line seemed impossibly far.

(2) He thought about quitting. His lungs ached; his side cramped. No one would blame him for slowing down—he was only in eighth grade, and this was the district final.

(3) Then he heard it: his little sister's voice in the crowd, shouting his name. She had watched him train every morning before dawn, timing his laps with their mother's old stopwatch.

(4) Something shifted. Diego lengthened his stride. One runner fell behind, then another. His legs found a rhythm he did not know he had.

(5) He did not win—he crossed the line in second place. But as he bent over, gasping, his sister threw her arms around him. "You passed two people!" she cried.

(6) Diego smiled through the pain. He had not come in first, but he had given everything he had. Some races, he realized, you win just by not giving up.

1. "The finish line seemed *impossibly far*" shows that Diego felt it was — A. easy to reach B. very hard to reach C. right in front of him D. behind him

2. What makes Diego push harder? — A. the coach B. his sister's voice C. the weather D. a trophy

3. "Something shifted" suggests that Diego — A. gave up B. found new motivation C. stopped running D. was injured

4. A theme of the story is that — A. winning is all that matters B. effort and perseverance matter C. running is easy D. crowds are loud

5. At the end, Diego's feelings are best described as — A. bitter B. proud despite not winning C. angry D. bored

6. Which line **best** shows Diego's determination? — A. "He thought about quitting" B. "Diego lengthened his stride" C. "the finish line seemed impossibly far" D. "his lungs ached"

7. "His legs found a rhythm" is an example of — A. simile B. personification C. hyperbole D. irony

SCR 1. *(2 points)* How does Diego change during the race? Support your answer with **evidence from the text.**

Passage 2 — The Clever Octopus (informational)

(1) The octopus may be the smartest animal without a backbone. These sea creatures can solve puzzles, use tools, and even recognize individual people.

(2) An octopus has about 500 million neurons—more than a rabbit. Strangely, two-thirds of them are not in its brain but in its eight arms. Each arm can taste, touch, and make decisions almost on its own.

(3) Octopuses are famous escape artists. In aquariums, they have opened jars, squeezed through tiny gaps, and sneaked out of tanks at night to hunt. One octopus learned to unscrew a jar lid to reach a crab inside.

(4) They also protect themselves cleverly. An octopus can change color and texture in an instant to blend into rocks or coral, and it can squirt ink to confuse a predator.

(5) Scientists are still learning how such a different kind of mind works. The octopus reminds us that intelligence can take many surprising forms.

8. The central idea of the passage is that octopuses are — A. dangerous B. surprisingly intelligent C. always in jars D. animals with backbones

9. As used in paragraph 3, "escape artists" suggests octopuses are good at — A. painting B. getting out of tight spaces C. swimming fast D. making ink

10. Paragraph 2 is mainly organized by — A. telling a story B. giving facts about the octopus's nervous system C. comparing colors D. listing dangers

11. Which detail **best** supports the idea that octopuses use tools? — A. "change color" B. "learned to unscrew a jar lid" C. "squirt ink" D. "500 million neurons"

12. From paragraph 2, you can infer that an octopus's arms — A. are useless B. can act with some independence C. cannot move D. are its only brain

13. The author's purpose is to — A. persuade you to buy an octopus B. inform readers about octopus intelligence C. tell a scary story D. sell aquariums

14. Which is the **best** summary? — A. Octopuses live in the sea. B. Octopuses are highly intelligent invertebrates that solve problems, use tools, and defend themselves cleverly. C. Octopuses have ink. D. Rabbits have neurons.

SCR 2. *(2 points)* Give **one** example that shows octopuses are intelligent. Use **evidence from the passage.**

Paired Set — Recycling

Text A — "Recycle Today" (persuasive)

Our town can do better. Right now, most of our trash ends up in a landfill, where it sits for decades. If we all recycled, we could cut that waste in half. Recycling is simple—start today, and make our community cleaner for everyone.

Text B — "Recycling by the Numbers"

Recycling one ton of paper saves about 17 trees and thousands of gallons of water. Towns with strong recycling programs send far less waste to landfills and often save money on trash disposal.

15. Text A is mainly written to — A. inform with statistics B. persuade people to recycle C. tell a story D. give directions

16. Text B mainly provides — A. an opinion B. facts and numbers about recycling's benefits C. a poem D. a complaint

17. Both texts agree that recycling — A. wastes time B. benefits the community C. is impossible D. harms trees

18. Which specific fact appears in Text B but **not** Text A? — A. recycling is good B. "17 trees" saved per ton of paper C. trash goes to a landfill D. we should start

19. "start today, and make our community cleaner" is an example of — A. a statistic B. a call to action C. a counter-argument D. a definition

20. Text B strengthens Text A mainly by — A. adding a story B. providing facts as evidence C. disagreeing with it D. using rhyme

Revising — "A Coding Club"

(1) Our school should start a coding club. (2) Students could learn to make games and apps. (3) Coding is a useful skill. (4) It could help with future jobs. (5) We would meet in the computer lab. (6) A coding club would be popular.

21. Which sentence could **best** be added after sentence 2 to develop it with a detail? — A. "Coding is fun." B. "For example, members could design a simple phone game by the end of the year." C. "I like computers." D. "Labs have computers."

22. Which transition would **best** begin sentence 4? — A. However, B. In addition, C. Because D. But

23. What is the **best** way to combine sentences 3 and 4? — A. Coding is useful, it helps with jobs. B. Coding is a useful skill that could help with future jobs. C. Useful skill, jobs, coding. D. Coding, useful, jobs.

24. Which sentence **best** states the writer's main idea (thesis)? — A. Sentence 1 B. Sentence 3 C. Sentence 5 D. Sentence 6

Editing — "The Garden"

(1) My grandmother grow tomatoes every summer. (2) Her and I water them each morning. (3) The plants grows taller every week. (4) Their is nothing like a fresh tomato.

25. What change should be made in sentence 1? — A. Change "grow" to "grows" B. Change "grandmother" to "grandmothers" C. Change "every" to "ever" D. No change

26. What change should be made in sentence 2? — A. Change "Her and I" to "She and I" B. Change "water" to "waters" C. Change "them" to "they" D. No change

27. What change should be made in sentence 3? — A. Change "grows" to "grow" B. Change "plants" to "plant" C. Change "taller" to "tall" D. No change

28. What change should be made in sentence 4? — A. Change "Their" to "There" B. Change "nothing" to "no thing" C. Change "fresh" to "Fresh" D. No change

Extended Constructed Response (ECR) — *(10 points)*

Text A, "Recycle Today," argues that the town should recycle more. **Should your community do more to encourage recycling?** Write a multi-paragraph **argumentative essay** that states your position and supports it with reasons and evidence, including evidence from the passages. Organize your essay with an introduction, body paragraphs, and a conclusion, and use correct grammar and spelling.

Answer key with TEKS tags for this bonus test is in the manuscript; score with the Quick Answer Keys method and the ECR/SCR rubrics.

Figure & Visual Practice

Many STAAR math questions include a **graph, diagram, or figure.** These items give you practice reading real visuals. (All figures below are exact, drawn-to-scale diagrams.) Answers are on the last page of this set.

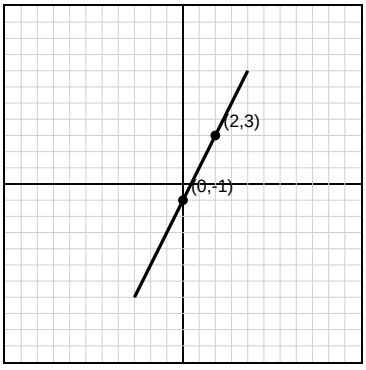
[IMAGE BOX — cover illustration]

Image prompt: "Clean, modern black-and-white book-cover graphic for a Texas STAAR Grade 8 study guide: a bold Texas star, an open book, and simple math/reading icons (ruler, pencil, speech bubble). Flat vector style, high contrast, no text."

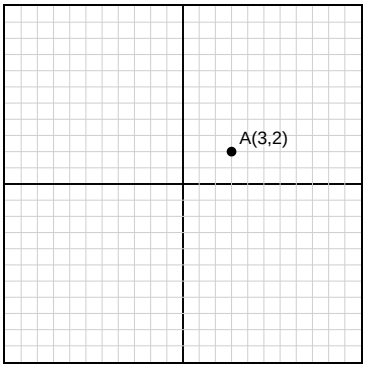
(Drop a real cover image here, or keep this prompt for a designer / image tool.)

Note: The box above shows how an illustration area is marked with an **image prompt** when a drawn diagram is not enough. The math figures in this book are exact diagrams, so they are drawn directly rather than generated.

Coordinate Grids

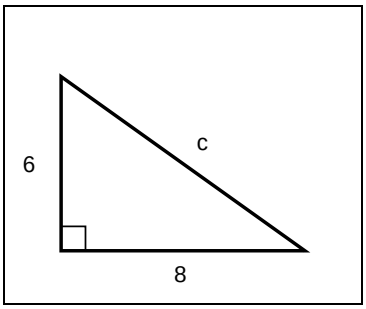


1. The line is graphed through the two marked points. What is its **slope**?
A. 1 B. 2 C. 3 D. 4

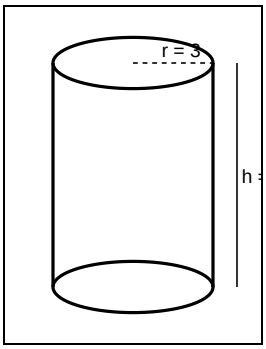


2. Point A is shown. What are the coordinates of its **reflection over the x-axis**? A. (−3, 2) B. (3, −2) C. (−3, −2) D. (2, 3)

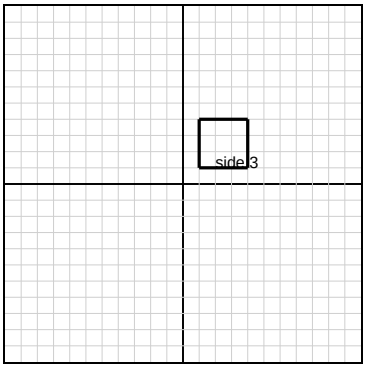
Geometry Figures



3. In the right triangle, the legs are 6 and 8. What is the length of the hypotenuse *c*? A. 10 B. 12 C. 14 D. 48

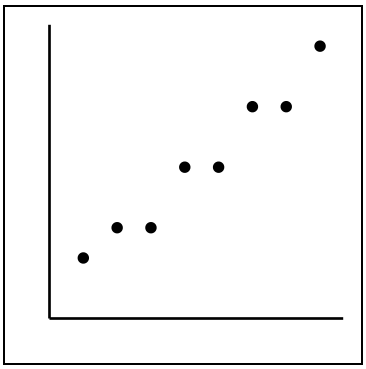


4. The cylinder has radius 3 and height 10. What is its **volume**? ($\pi \approx 3.14$)
A. 94.2 B. 188.4 C. 282.6 D. 565.2

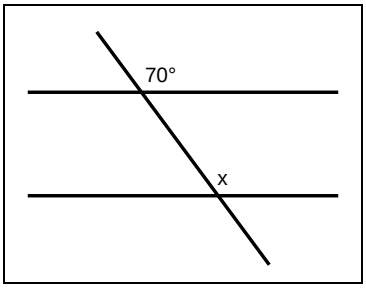


8. The square shown has a side of 3. If it is dilated by a scale factor of 2, what is the **area of the image**? A. 12 B. 18 C. 24 D. 36

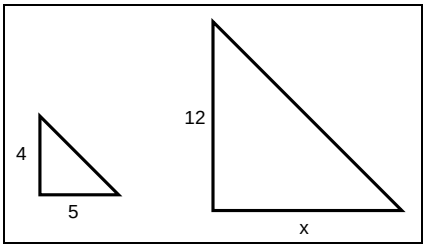
Data & Angles



5. The scatterplot shows the relationship between two variables. What **association** does it show? A. Positive linear B. Negative linear C. No association D. Non-linear



6. Two parallel lines are cut by a transversal. One angle measures 70°. What is the measure of angle *x* (its corresponding angle)? A. 20° B. 70° C. 110° D. 180°



7. The two triangles are similar. The small triangle has sides 4 and 5; on the large triangle the side matching the 4 is 12. What is the length *x* that matches the 5? A. 13 B. 14 C. 15 D. 20

Figure Practice — Answer Key

1. B *(8.4C ★)* — From (0,−1) to (2,3): rise 4, run 2 → slope = 4/2 = **2**.

2. B *(8.10C ★)* — Reflecting A(3,2) over the x-axis flips the y-sign → **(3, −2)**.

3. A *(8.7C ★)* — $c^2 = 6^2 + 8^2 = 36 + 64 = 100$ → c = **10**.

4. C *(8.7A ★)* — $V = \pi r^2 h = 3.14 \times 9 \times 10 =$ **282.6**.

5. A *(8.11A ○)* — The points rise left to right → **positive linear** association.

6. B *(8.8D ○)* — Corresponding angles formed by a transversal are equal → **70°**.

7. C *(8.3A ○)* — Scale factor = 12/4 = 3, so x = 5 × 3 = **15**.

8. D *(8.10D ○)* — Area scales by k^2: original area 3×3 = 9, image = $9 \times 2^2 =$ **36**.

Final Notes

You Are Ready

If you have worked through the lessons, taken the practice tests, reviewed the explanations, and used the diagnostic charts, you have done the hard, honest work that builds real confidence. You have practiced every skill the STAAR Grade 8 Reading Language Arts and Mathematics tests can ask of you—reading closely, reasoning with evidence, and solving real problems.

On test day, trust your preparation. Read carefully, answer every question, give the written responses your best effort, and use your full time. You've got this.

A Word About What Comes Next

Spring 2027 is the final STAAR administration for grades 3–8. A new Texas assessment will follow, but the skills you have strengthened here do not expire. Strong readers and problem-solvers succeed on any test—and far beyond it.

About the Series

This **2027 Edition** is part of a Grade-by-Grade Texas test-preparation series created to give students clear teaching, realistic practice, and honest feedback—100% aligned to the Texas Essential Knowledge and Skills, with nothing wasted.

Author: Andrian Ross

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Content reflects the STAAR program as of **November 5, 2026**. Test specifications can change; confirm current details at the official Texas Education Agency website.